SCALE-COVARIANT ATTENTION FOR TRAINING-FREE HIGH-RESOLUTION DIFFUSION TRANSFORMERS

Anonymous authors
Paper under double-blind review

ABSTRACT

Diffusion transformers trained near one megapixel fail at higher resolutions: attention meets unseen token counts and position offsets, and its cost grows quadratically. Each head's balance between local detail and global layout, we find, is measurable at the training resolution and transfers to larger images. Scale-Covariant Attention (SCA) calibrates once, fitting one weight per head and noise level in closed form against the model's full attention, and reuses that table at the target. Each head mixes a native-size sliding window with block-centre global attention, both in the trained position range, after a short global-only opening that sets the layout. For FLUX.1-dev at 4096^2, SCA is the strongest single-pass training-free operator we tested: it beats every shared-sampler baseline on all per-image metrics, has the best FID and fine detail of eight methods, and runs $2.85\times$ faster than AReA. Dropped into HRDiT's progressive schedule, it matches HRDiT's preference scores in less than half the time. Gains hold at 6144^2, with two more seeds, and on FLUX.2-dev; on FLUX.1-Krea-dev SCA matches AReA.

1 INTRODUCTION

Diffusion transformers (Peebles & Xie, 2022) such as FLUX.1 (Black Forest Labs, 2024) and Stable Diffusion 3 (Esser et al., 2024) are trained at about one megapixel. At 4096×4096 the image-token count grows 16-fold and full-attention cost about 256-fold, and every attention layer sees token counts and position offsets outside its training distribution. Outputs show duplicated subjects, broken layouts, and blurred texture (He et al., 2023; Du et al., 2023; Issachar et al., 2025; Rajabi et al., 2026a); in FLUX.1-dev at 4096^2, full attention collapses into stripes or a flat field (Figure 1).

Training-free fixes edit the positional encoding (Zhuo et al., 2024; Issachar et al., 2025; Zhao et al., 2025; Rajabi et al., 2026b) or restrict the keys each query sees (Zhang et al., 2023; Ma et al., 2025; Xue et al., 2026; Rajabi et al., 2026a). AReA (Rajabi et al., 2026a), the most recent of the second kind, attends to keys subsampled onto a native-size grid for the first 8 of 28 steps, then to windows that shrink to native size, and shares this schedule across heads. Head-specific receptive fields are also established: HRDiT assigns per-head scopes under a cost budget (Xue et al., 2026), and UltraImage concentrates attention per head using entropy (Zhao et al., 2025). We ask whether a head's local/global output mixture can be fitted at native resolution and transferred to larger images.

We answer it by emulation. To mimic a target s times larger, we shrink both a native-size neighbourhood and a downsampled view of the whole image by s, so the image-to-window ratio matches the target while full attention stays in distribution and serves as ground truth. A convex mix of the two views fits each head well. The fitted weight differs strongly between heads, is stable across tested prompt splits (correlation 0.99), and transfers: the table fitted at emulated scale 2 explains scale 4 with 2.4% more output error than the scale-4 table, and a 1024^2 table explains 1280^2 with 0.8% more (Section 3). We call this measured transfer scale covariance: the local view follows a law fixed in tokens, the global view a law fixed in image fractions, and over the tested scales the weight between them stays nearly constant.

Scale-Covariant Attention (SCA) turns this measurement into an attention operator (Figure 2). Each head uses its calibrated weight to mix two branches: a global branch that attends to block-centre keys on the native grid with rescaled coordinates, and a local branch that is an exact native-size sliding window. The calibrated mix alone does not preserve layout: applied from the first step at

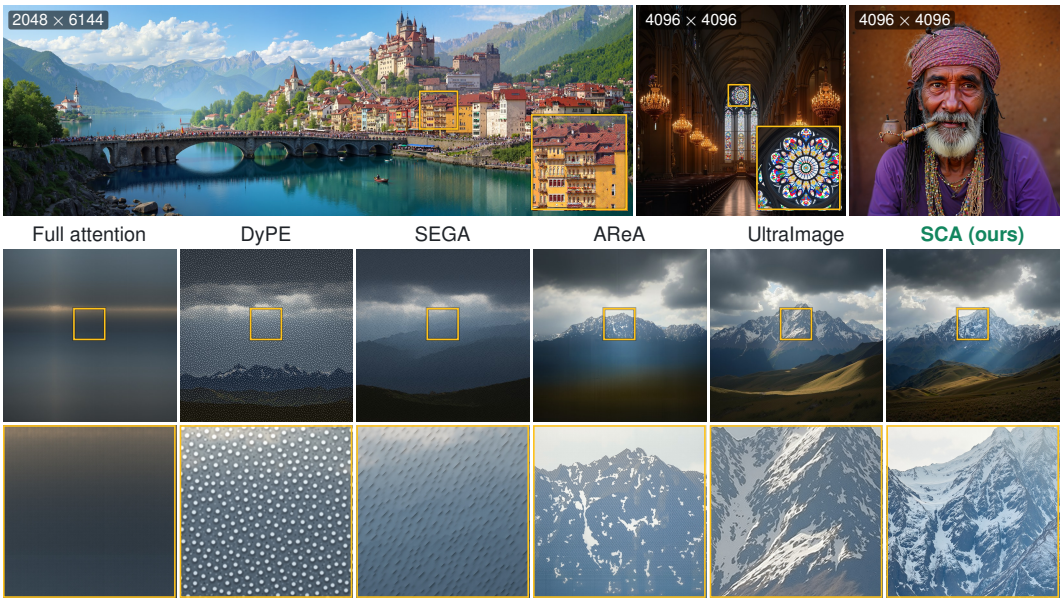

Figure 1: **SCA keeps layout and fine detail at 4096^2 and 2048×6144 without training, $2.85\times$ faster than AReA.** Top: curated SCA outputs with native-pixel insets. Bottom: one curated prompt at 4096^2, each full image above the same 1:1 crop; UltraImage uses its own two-stage recipe.

4096^2, it draws five Moai statues for a prompt asking for one (Figure 4). SCA therefore runs the global branch alone for the first 8 of 28 steps (the opening phase), as a frequency argument suggests (Argument 1). Our contributions are:

- **Calibration by emulation.** A closed-form procedure that emulates the target scale inside the training distribution and fits each head's receptive-field mix to the model's own full attention from one pass of statistics (Proposition 3). On FLUX.1-dev the mix is head-specific, prompt-stable, and transfers across scale and calibration resolution.
- **The SCA operator.** A per-head, per-noise-level mix of two branches whose image query-key offsets stay in the trained range (Proposition 1) at a cost linear in the token count (Proposition 2). One attention layer at 4096^2 takes 18.7 ms, full attention 87.3 ms.
- **Paired evidence against seven baselines.** Full attention, AReA, DyPE, SEGA, and SigMa under one FLUX.1-dev protocol, and UltraImage and HRDiT under their own multi-stage recipes, with paired bootstrap intervals, ablations, failure galleries, and checks on HRGen-1K, extra seeds, and FLUX.2-dev. Against AReA on 191 prompts at 4096^2, SCA wins 11 of 15 metrics with paired 95% intervals excluding zero, loses none, and runs $2.85\times$ faster.
- **A drop-in operator for progressive sampling.** Replacing all four of HRDiT's attention changes inside its schedule, SCA matches HRDiT's preference scores (no per-image metric significantly worse; HPSv2 and ARNIQA significantly better) in less than half the time.

2 Scale-Covariant Attention

2.1 Setting and Notation

A multimodal diffusion transformer such as FLUX.1 concatenates T text tokens $\mathcal{T}$ with $N = HW$ image tokens on the grid $\mathcal{G} = [H] \times [W]$, $[n] = \{0, \dots, n-1\}$, and applies joint attention in each block (57 blocks of 24 heads in FLUX.1). It is trained at about 1024^2 pixels, a native grid of $n_0 \times n_0$ tokens with $n_0 = 64$. Rotary embeddings (Su et al., 2021) give token j a position $p_j \in \mathbb{Z}^2$ (FLUX.1 puts all text tokens at 0) and a rotation R_p with $R_p^\top R_{p'} = R_{p'-p}$. For one head, a pre-rotation query

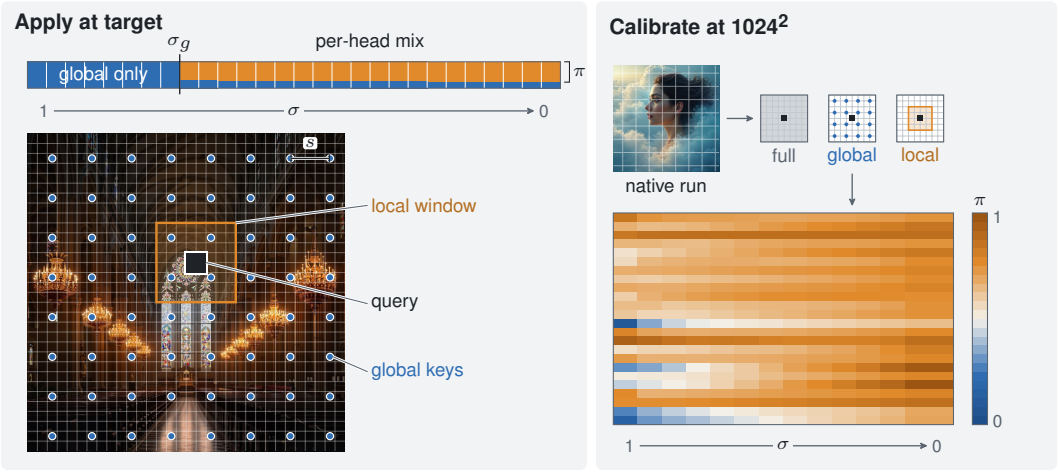

Figure 2: **SCA mixes a native-size local window with block-centre global keys.** The opening steps, $\sigma > \sigma_g$, use the global operator alone, since high-noise windows repeat structure. Afterwards each head weights the local output by π, fitted once at 1024^2 by matching full attention (right: one block's 24 heads by 14 noise bins).

q placed at u, and a set $\mathcal{K}$ of (key, value, key position) triples,

$$\mathcal{A}(q, u; \mathcal{K}) \;=\; \textstyle\sum_{(k,v,u')\in\mathcal{K}} \operatorname{softmax}_{\mathcal{K}}\big(q^\top R_{u'-u}\, k/\sqrt{d}\big)\; v. \tag{1}$$

Full attention is $O_i^{\text{full}} = \mathcal{A}\big(q_i, p_i; \{(k_j, v_j, p_j)\}_{j\in\mathcal{T}\cup\mathcal{G}}\big)$. The target grid is $H = s_H n_0$, $W = s_W n_0$ with integer factors (s when equal), so full attention meets offsets s times the trained range. The latent at noise level σ is $x_\sigma = (1-\sigma)x_0 + \sigma\epsilon$. Splitting the keys into a near and a far set gives exactly $O_i^{\text{full}} = m_i O_i^{\text{near}} + (1-m_i) O_i^{\text{far}}$, where O_i^{near} and O_i^{far} attend only to their own set and m_i is the softmax mass on the near set (Lemma 1): every head is a query-dependent mix of a near and a far field. SCA approximates this mix with two separately normalized operators and one weight per head. Because both branches include text keys and their image key sets can overlap, the identity does not decompose SCA exactly.

2.2 Two operators with native-range image geometry

Global operator. Let $\mathcal{G}_0 = [H/s_H] \times [W/s_W]$ be the $n_0 \times n_0$ grid of blocks. Block-centre subsampling keeps for each block b the token at $\kappa(b) = (s_H b_1 + \lfloor s_H/2 \rfloor,\; s_W b_2 + \lfloor s_W/2 \rfloor)$, and the coarse map $c(p) = (\lfloor p_1/s_H \rfloor, \lfloor p_2/s_W \rfloor)$ sends a position to its block, so $c(\kappa(b)) = b$:

$$\mathcal{K}^{\text{glob}} = \{(k_j, v_j, 0)\}_{j\in\mathcal{T}} \cup \{(k_{\kappa(b)}, v_{\kappa(b)}, b)\}_{b\in\mathcal{G}_0}, \qquad O_i^{\text{glob}} = \mathcal{A}\big(q_i,\, c(p_i);\, \mathcal{K}^{\text{glob}}\big), \tag{2}$$

with $c(p_i)$ replaced by 0 for text queries. This is AReA's first phase with the pooling factor fixed at s. It models content whose extent is a fixed fraction of the image: layout, object count, identity.

Local operator. Image queries form tiles of $t \times t$ tokens ($t = 8$); query i lies in tile $\tau = \lfloor p_i/t \rfloor$. With window side $w = \min(n_0, H)$, the window of tile τ starts at $a(\tau) = \operatorname{clip}\big(t\tau + \lfloor t/2 \rfloor - \lfloor w/2 \rfloor, 0, H - w\big)$ per axis, centred on the tile and clamped to the grid. With $\mathcal{W}(i)$ the $w \times w$ window of i's tile,

$$\mathcal{K}_i^{\text{loc}} = \{(k_j, v_j, 0)\}_{j\in\mathcal{T}} \cup \{(k_j, v_j, p_j)\}_{p_j\in\mathcal{W}(i)}, \qquad O_i^{\text{loc}} = \mathcal{A}\big(q_i,\, p_i;\, \mathcal{K}_i^{\text{loc}}\big), \tag{3}$$

an exact sliding window with the original positions. It models content whose extent is fixed in tokens: edges, texture, fine structure.

Proposition 1 (Trained position range)**.** *Assume exact integer position coordinates and $t \le n_0 \le \min(H, W)$. Every query-key offset of the global operator and every image query-key offset of the local operator is at most $n_0 - 1$ per axis, and each image query sees $T + n_0^2$ keys in either operator, as in full attention on the native grid.*

Proposition 2 (Cost). *Per head and layer, full attention scores $(N+T)^2$ query-key pairs, the global operator $(N+T)(T+N_0)$ with $N_0 = |\mathcal{G}_0| = n_0^2$, and the local operator $N(T+w^2)$. For fixed n_0 and T the total is $O\big(N(T+w^2+N_0)\big)$, linear in N.*

Proofs are in Appendix A. Proposition 1 assumes exact coordinates; the frozen FLUX.1 implementation rounds positions in bf16 beyond 256 tokens per axis (Appendix C). Native-range geometry also does not make hidden states in-distribution, and text-image offsets in the local operator equal p_i, as in full attention at the target (Remark 1). At 4096^2 SCA scores 7.2 times fewer query-key pairs than full attention, 8.4 times fewer with the opening phase (Table 4).

2.3 Per-head mixing and the opening phase

For an image query of head h in block l at noise level σ,

$$O_i \;=\; O_i^{\mathrm{glob}} \;\text{ if } \sigma > \sigma_g, \qquad O_i \;=\; \pi_{lh}(\sigma)\, O_i^{\mathrm{loc}} + \big(1-\pi_{lh}(\sigma)\big)\, O_i^{\mathrm{glob}} \;\text{ otherwise,} \tag{4}$$

and text queries use O_i^{glob}. The weight $\pi_{lh}(\sigma)$, written $\pi_h(\sigma)$, stands in for the near-field mass m_i; one weight per head is cheap and a few prompts determine it.

Argument 1 (A frequency-based explanation for the opening phase). At noise level σ a spatial frequency band has signal-to-noise ratio $\big((1-\sigma)/\sigma\big)^2 S(f)$ for image spectrum S. If latent power falls with frequency, only low frequencies carry signal at high noise, and a window of w tokens spans less than one period of any frequency below $1/w$. Independently attended windows may then repeat local structures instead of forming one large object, while a global branch supplies context across windows. Hidden states also carry information from earlier layers, so this is a mechanistic hypothesis, not an impossibility result (Appendix A).

Output matching cannot see this failure: a tokenwise squared error can favour local detail without preserving layout. We have not measured per-band signal-to-noise ratios of FLUX.1 latents, but a 16-token window gives smaller and more numerous copies than a 64-token one, as predicted (Figure 4). We set $\sigma_g = 0.89$ (the first 8 of 28 steps at maximum shift, as in AReA); gating 13 steps gives a 4×4 block mosaic, because all queries of a block share one coarse position.

2.4 Calibration by output matching

The weights must come from where the model is trustworthy, which the target is not. On an in-distribution $n \times n$ grid ($n = 64$ at 1024^2), emulated scale s_c uses subsampling factor s_c, window side n/s_c, and tile side $\mathrm{round}(tn/(s_c n_0))$, so that the ratio of grid to window is s_c, as at a target of scale s_c. The sampler runs unmodified full attention; at every block and step both emulated operators are evaluated on the same queries, keys, and values. For block l, head h, and noise bin β, sums over prompts, steps with $\sigma \in \beta$, image queries, and channels give

$$A = \sum \|O^{\mathrm{loc}} - O^{\mathrm{glob}}\|^2, \quad B = \sum \langle O^{\mathrm{full}} - O^{\mathrm{glob}},\, O^{\mathrm{loc}} - O^{\mathrm{glob}} \rangle, \quad E = \sum \|O^{\mathrm{full}} - O^{\mathrm{glob}}\|^2, \tag{5}$$

and the squared error of the mix with weight π is $J(\pi) = E - 2\pi B + \pi^2 A$.

Proposition 3 (Closed-form calibration and transfer cost). *In each cell (l, h, β), J is minimized on $[0,1]$ by $\pi^\star = \mathrm{clip}(B/A, 0, 1)$, with $\pi^\star = 1$ if $A = 0$. Any other weight $\pi' \in [0,1]$ (fitted at another scale or resolution, per layer, shared, or a hard switch) has $J(\pi') - J(\pi^\star) \geq A(\pi' - \pi^\star)^2$, with equality when $0 < \pi^\star < 1$.*

So the per-head table has the lowest output error of any rule constant within a cell, and replacing it costs an A-weighted squared distance. This holds at the emulation scale; Section 3 measures how well tables transfer between scales and resolutions. The same sums score every alternative rule without another forward pass. We use 14 noise bins of width $1/14$ and interpolate linearly between the mean noise levels of occupied bins at inference. One pass collects emulated scales 2 and 4. A target with factors (s_H, s_W) uses the table whose scale is nearest $\max(s_H, s_W)$ in log space, here always 4. The local operator is one FlexAttention call (Dong et al., 2024) over tile-major tokens with the window as an exact block mask (Algorithm 1, Appendix C).

2.5 SCA INSIDE A PROGRESSIVE SCHEDULE

SCA replaces only the attention, so it can serve inside a multi-stage sampler. HRDiT (Xue et al., 2026) denoises at 1024^2 for 30 steps, then upsamples, re-noises, and repeats the last 17 and 10 steps of the same schedule at 2048^2 and 4096^2, pulling each estimate's low frequencies toward the previous stage. At those two stages it changes attention in four ways: NTK-scaled RoPE, a proportional attention scale, per-head key pruning, and position alignment. Progressive SCA keeps HRDiT's schedule, transitions, guidance, and seeds, and replaces all four changes with SCA: native RoPE with the scale-2 table at 2048^2 and the scale-4 table at 4096^2. The stages re-enter at $\sigma = 0.81$ and 0.61, below $\sigma_g = 0.89$, so the opening phase never runs; the 1024^2 stage sets the layout. All settings were fixed before any evaluation image was generated (Appendix G).

3 WHAT THE HEADS WANT

We calibrate FLUX.1-dev on 32 purpose-written prompts at 1024^2 and 16 at 1280^2, disjoint from the evaluation sets. Errors are relative, $\sum J / \sum \|O^{\text{full}}\|^2$, pooled over blocks, heads, and steps.

Heads differ. Over the 1368 heads, the standard deviation of $\pi_h(\sigma)$ is 0.24 to 0.26 at every noise level (emulated scale 2). Half of the (head, noise level) cells lie in $[0.2, 0.8]$, 11.7% exceed 0.95, and 2.9% are below 0.05 (Figure 2a; full tables in Figure 5, Appendix B). The mean weight is 0.64 to 0.74, so the typical head leans local but still draws on the global branch.

The table is stable across tested prompt splits. Tables fitted on disjoint halves of the prompts correlate at 0.989 over all cells (0.986 at scale 4; 0.974 for 16 prompts at 1280^2), and each half's table scored on the other half gives a relative error of 0.0576 against 0.0574 in sample. Tables from 4, 8, or 16 prompts also generate similarly (Section 4.3).

The weights transfer across the tested scales and resolutions. The scale-2 table applied at scale 4 gives a relative error of 0.205 against 0.200 for the scale-4 table, still below every shared rule fitted at scale 4. The 1024^2 table scored at 1280^2 gives 0.0589 against 0.0585. Over all pairs of calibration targets, a transferred table adds at most 5.45% error, while each target's own per-layer fit adds 12 to 24%. The two resolutions emulate scale 2 with windows of 32 and 40 tokens, so this transfer is consistent with weights that depend mainly on the image-to-window ratio.

Per-head beats coarser rules; output error misses layout. Relative to the best hard switch, per-head mixing lowers the relative error from 0.092 to 0.057 at scale 2 and from 0.261 to 0.200 at scale 4; per-layer tables reach 0.071 and 0.223, one shared weight per noise level 0.077 and 0.230 (Figure 6). In output energy the best hard switch never switches to global (Appendix B), so the opening phase deliberately overrides the fit (Argument 1).

4 EXPERIMENTS

Protocol. All FLUX.1-dev comparisons share AReA's sampler: 28 Euler flow-matching steps, guidance 4.5, shift $\mu = 1.15$ at every size, bf16, tiled VAE decoding, and seed label 0 for every prompt. The main prompts are the Aesthetic-4K evaluation split with 4K references (Zhang et al., 2025a): 191 prompts after excluding four development prompts, with a fixed 46-prompt subset for ablations. Sizes, written height×width, are 4096^2, 2048×4096, and 2048×6144 (191 prompts each), 6144^2 (100 prompts), and 8192^2 (12 exploratory prompts). Baselines are full attention, AReA from its released code (Appendix F), DyPE (Issachar et al., 2025), SEGA (Rajabi et al., 2026b), UltraImage (Zhao et al., 2025), and HRDiT (Xue et al., 2026) on all 191 prompts, plus SigMa (Zhao et al., 2026) on the 46-prompt subset; this fixes the sampler rather than each baseline's tuned defaults (Appendix G). UltraImage runs its released guided two-stage recipe (30 steps per stage, guidance 3.5 then 7) and HRDiT its progressive recipe (30, 17, and 10 steps at 1024^2, 2048^2, and 4096^2); both are system-level comparisons with their own budgets. We report preference and alignment scores (ImageReward, HPSv2/2.1, PickScore, CLIP, aesthetic; Xu et al., 2023; Wu et al., 2023; Kirstain et al., 2023; Radford et al., 2021; Schuhmann, 2022), distribution scores (FID, KID, patch FID/KID, Inception score; Heusel et al., 2017; Salimans et al., 2016), and no-reference quality (MUSIQ on the

Table 1: **Among single-pass methods, SCA has the best mean on every metric shown, and it is the fastest.** 4096^2, 191 Aesthetic-4K prompts, FLUX.1-dev. HPSv2.1 $\times 100$; MUSIQ_c: MUSIQ on 1024^2 crops; FID_p: patch FID. **Bold** / underline: best / second best. Seconds per image on one GH200: clean: batch-1 median of three images after warm-up (for HRDiT, including its cold first run); camp.: campaign mean on shared nodes; n/m: not measured. All metrics: Table 8.

		Time (s)↓		Image		Pick				
Method	Steps	clean	camp.	Reward	HPSv2.1	Score	CLIP	MUSIQ_c	FID↓	FID_p↓
Single pass, shared sampler (28 steps, guidance 4.5)										
Full attention	28	186.5	196.5	−2.18	7.61	17.94	15.38	33.2	312.7	252.4
DyPE	28	n/m	196.1	−0.11	20.52	20.49	24.84	50.4	206.2	172.7
SEGA	28	n/m	190.5	0.55	24.37	21.64	26.93	50.4	165.4	126.0
AReA	28	202.4	261.7	0.92	27.20	22.35	27.67	50.4	146.0	103.4
SCA (ours)	28	**71.1**	**71.5**	1.23	29.71	23.09	**29.11**	**61.2**	**138.4**	**99.8**
Multi-stage, own recipe and budget										
UltraImage	30+30	n/m	308.5	1.28	30.80	23.32	28.80	41.8	142.7	110.6
HRDiT	30+17+10	108.6	110.2	**1.34**	**30.96**	**23.45**	29.10	53.5	139.1	103.9

Table 2: **On a 46-prompt panel, SCA leads the single-pass methods, and in HRDiT's schedule it matches HRDiT's preference scores in less than half the time.** Columns as in Table 1; campaign times from this panel. Against SCA at 28 steps (Tables 21, 22, and 23): $^+$ significantly better, $^\circ$ 95% paired interval includes zero, unmarked significantly worse. **Bold**: best. 46-image FID is exploratory; budget runs lack whole-image FID (n/a); *: patch FID without interval. 42 steps: sized to HRDiT's campaign time; progressive: HRDiT's schedule with SCA attention (Section 2.5).

		Time (s)↓		Image		Pick				
Method	Steps	clean	camp.	Reward	HPSv2.1	Score	CLIP	MUSIQ_c	FID↓	FID_p↓
Single pass, shared sampler (28 steps, guidance 4.5)										
Full attention	28	186.5	191.6	−2.21	7.00	17.73	14.89	33.6	370.1	292.4
DyPE	28	n/m	204.2	−0.23	20.24	20.25	24.86	50.9	282.4	251.5
SigMa	28	n/m	185.5	−0.77	18.16	19.67	22.95	50.4	289.5	255.4
SEGA	28	n/m	189.7	0.47	24.82	21.56	27.36	51.3	243.4	198.3
AReA	28	202.4	241.0	0.83	27.76	22.31	27.85	52.0	212.7°	168.4°
SCA (ours)	28	**71.1**	71.6	1.12	29.98	22.96	**29.41**	62.5	**206.4**	168.0
Multi-stage, own recipe and budget										
UltraImage	30+30	n/m	322.0	1.26°	31.54$^+$	23.31$^+$	28.77	42.9	213.8°	182.2
HRDiT	30+17+10	108.6	106.5	**1.37**$^+$	31.60$^+$	**23.42**$^+$	29.38°	54.1	216.9°	178.1
SCA with other budgets or schedules										
56 steps	56	n/m	140.1	1.22$^+$	30.99$^+$	23.12$^+$	29.09°	**64.7**$^+$	n/a	175.1
guidance 7	28	n/m	71.9	1.14°	29.24	22.69	28.98°	63.4°	n/a	**167.2***
42 steps	42	n/m	110.4	1.19°	30.62$^+$	23.02°	29.12°	64.0$^+$	209.2°	170.3°
progressive	30+17+10	n/m	**49.0**	1.30$^+$	**31.88**$^+$	23.41$^+$	29.02°	54.7	213.4	181.7

full image and on 1024^2 crops, MUSIQ_c; CLIP-IQA; ARNIQA; Ke et al., 2021; Wang et al., 2022; Agnolucci et al., 2024), with 95% paired bootstrap intervals over prompts. SCA uses $t = 8$, $w = 64$, $\sigma_g = 0.89$, and the scale-4 table from 32 prompts at 1024^2. Each process uses one NVIDIA GH200 120GB. Table 7 and Appendices D and E define every setting and metric.

4.1 Main results

Against AReA. At 4096^2, SCA beats AReA on every per-image metric at $2.85\times$ the speed (Table 1, Figures 1 and 3). Eleven of 15 metrics improve significantly, including ImageReward ($+0.31$ $[0.24, 0.38]$) and FID (-7.6 $[-12.8, -2.9]$), and the other four are unresolved (Table 9).

Other baselines. SCA also beats DyPE and SEGA on every per-image metric over 191 prompts (Figure 10), and all five shared-sampler baselines on the 46-prompt panel (Table 21). UltraImage,

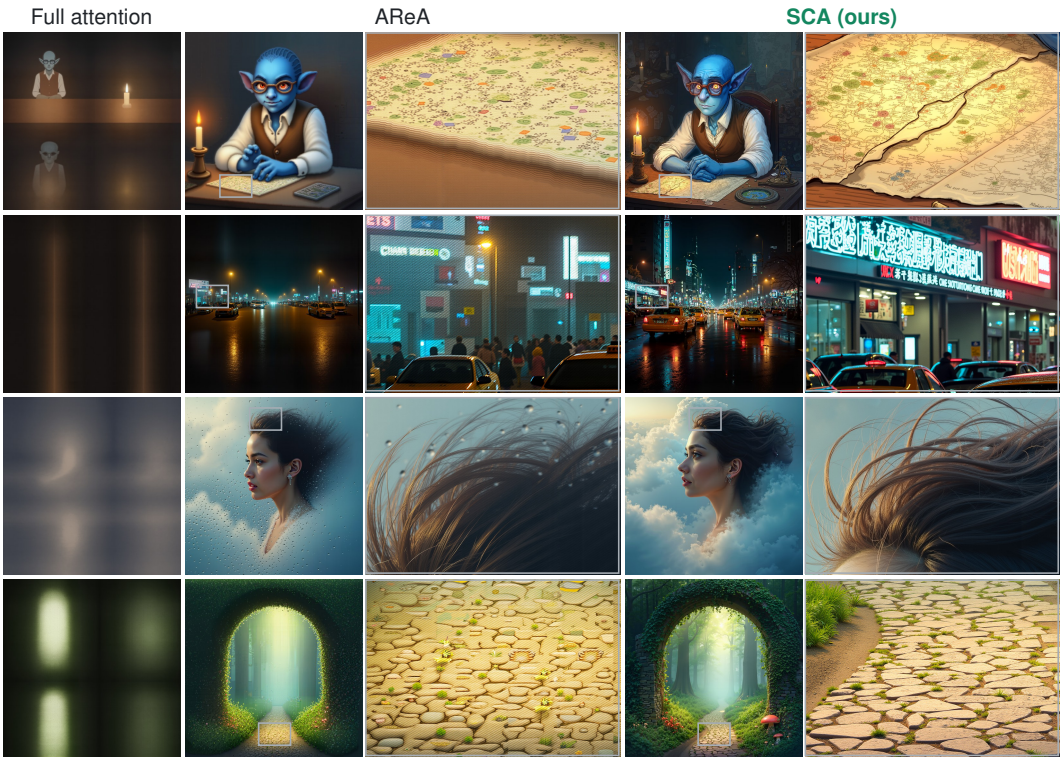

Figure 3: **At 4096^2 full attention collapses, AReA blurs or tiles fine structure, and SCA keeps both layout and detail.** Four curated prompts, seed 0, picked for variety from the 191 evaluation prompts. Beside each AReA and SCA image is a 1:1 crop of the same 736×512 window (placement: Appendix P). Uncurated samples: Figure 14.

at over four times SCA's time, is significantly better on five per-image metrics, including HPSv2.1 and PickScore, and worse on CLIP score and both MUSIQ scores, with ImageReward unresolved and higher FID (Tables 17 and 20).

HRDiT and progressive SCA. HRDiT's three-stage recipe leads single-pass SCA on learned preference scores at $1.5\times$ SCA's time (Table 1). The lead comes from HRDiT's schedule: progressive SCA, that schedule with SCA attention, matches HRDiT's preference scores in less than half the time (Table 2; Figures 11 and 12). ImageReward differs by -0.08 $[-0.20, 0.03]$, HPSv2.1 by $+0.28$ $[-0.21, 0.77]$, and PickScore by -0.02 $[-0.18, 0.14]$; no per-image metric is significantly worse, HPSv2 and ARNIQA are significantly better, and none of this establishes equivalence (Table 23). Spending HRDiT's time on 42 single-pass steps does not close the gap. The parity costs fidelity: progressive SCA has significantly higher FID and patch FID and lower MUSIQ_c than single-pass SCA, which keeps the fidelity lead (Table 2, Appendix J).

Other sizes. Gains depend on the aspect ratio (Tables 10 and 16): SCA beats AReA on 14 of 15 metrics at 6144^2 and on ten at 2048×6144, where it loses on aesthetic score, but results are mixed at 2048×4096. At 8192^2, 12 exploratory prompts favour SCA on preference and alignment scores, though grid artifacts remain (Appendix K).

Latency. SCA is the fastest method we timed at every size, and its time grows nearly linearly with pixel count (Figure 9, Table 5). At 6144^2 it takes 161.6 s per image, $4.7\times$ faster than full attention and $3.0\times$ faster than AReA.

Table 3: **Every ablation significantly lowers HPSv2.1 and PickScore.** 4096^2, 46 prompts. First row: SCA's absolute values; other rows: paired difference after changing one component, in **bold blue** / orange when significantly better / worse (95% interval excludes zero). HPSv2.1 and CLIP-IQA are $\times 100$. Mix err.: relative output error at emulated scale 4. Full results: Table 11.

Variant	Mix err.	ImageReward	HPSv2.1	PickScore	CLIP-IQA	MUSIQ_c
SCA (reference)	0.200	1.12	29.98	22.96	71.3	62.5
Weight rule after the opening phase						
local only ($\pi = 1$)	0.261	−0.36	−2.86	−1.20	−4.6	−0.3
one shared $\pi(\sigma)$	0.230	−0.09	−1.83	−0.38	−7.7	−10.4
per-layer $\pi_\ell(\sigma)$	0.223	−0.05	−0.70	−0.16	−0.9	−2.4
Global branch and opening phase						
global only ($\pi = 0$)	0.627	−0.40	−5.66	−1.44	−27.0	−11.7
mean pooling	n/a	−1.52	−9.75	−2.60	−10.1	−4.2
no opening phase (v0)	0.200	−1.33	−4.07	−2.18	**+2.9**	+2.5

4.2 Ablations

Table 3 changes one component at a time (examples in Figure 19). **The opening phase is essential.** Without it (v0), ImageReward falls by 1.33 $[1.06, 1.60]$ and subjects repeat; CLIP-IQA and ARNIQA rise because each copy is sharp. **The per-head mix carries global context after the opening phase.** Local-only attention in the last 20 steps lowers ImageReward by 0.36 $[0.21, 0.52]$, HPSv2.1 by 2.86, PickScore by 1.20, and CLIP score by 1.93. **Per-head weights beat shared and per-layer weights, in the order output error predicts.** One shared weight per noise level (mix error 0.230) lowers ImageReward by 0.09 $[0.01, 0.17]$ and MUSIQ_c by 10.4; per-layer weights (0.223) still lose significantly on HPSv2.1, PickScore, ARNIQA, and MUSIQ_c. **Neither branch works alone, and block centres beat block means.** Global only loses on every metric, and mean pooling blurs the global keys (ImageReward -1.52). Band-guard ablations, including hard per-head routing, are in Appendix M.

4.3 Generality and sensitivity

SCA's gains over AReA hold with seeds 1 and 2 and on 200 HRGen-1K prompts (Rajabi et al., 2026a), where all ten per-image metrics improve. They barely depend on the calibration set: 4 to 16 prompts instead of 32 change ImageReward by at most 0.012 (Table 15). Moving the gate to 0.85 or 0.93 loses on some metrics. On FLUX.2-dev, SCA raises ImageReward over full attention from 0.75 to 1.20 in half the time. On FLUX.1-Krea-dev, in a third of the time, it wins six per-image metrics, leaves ImageReward unresolved, and has higher patch FID (Table 16).

5 Related work

Training-free high-resolution sampling. Training-free methods alter convolutions, spectra, or attention temperature (He et al., 2023; Zhang et al., 2023; Huang et al., 2024; Jin et al., 2023), fuse global and patch paths (Du et al., 2023; Yang et al., 2024; Qiu et al., 2024), add native-resolution guidance (Du et al., 2024; Bu et al., 2025; Koh et al., 2025), or adapt RoPE (Zhuo et al., 2024; Issachar et al., 2025; Zhao et al., 2026; Rajabi et al., 2026b; Liu et al., 2026). Closest to SCA, AReA shares one schedule of pooled global attention and shrinking windows across heads (Rajabi et al., 2026a); ResDiT spectrally fuses position-scaled global and native-size patch attention between global-only and local-only phases (Ma et al., 2025); and FreeScale fuses attention outputs by frequency (Qiu et al., 2024).

Head- and layer-specific attention. HRDiT picks per-head scopes by loss sensitivity under a cost budget (Xue et al., 2026), and UltraImage concentrates attention per head by entropy, noting stable local/global head roles before us (Zhao et al., 2025). Sparse attention in diffusion transformers (Yuan et al., 2024; Xi et al., 2025; Zhang et al., 2025b), language models (Fu et al., 2024; Xiao et al., 2024), and vision transformers (Pan et al., 2022) varies windows, spans, or key scales across layers

or heads. Our local branch is neighbourhood attention (Hassani et al., 2022), and ToDo attends to downsampled keys without rescaling coordinates (Smith et al., 2024).

Head specialization, two branches, and a global opening phase thus all have precedents; SCA adds calibration by emulation, a closed-form per-head, per-noise-level mixture fitted at native resolution and transferred to an operator with native key counts.

6 Limitations and conclusion

Limitations. Multi-stage recipes keep higher learned preference scores than single-pass SCA, HRDiT at $1.5\times$ and UltraImage at over four times its time; progressive SCA closes the gap to HRDiT but costs fidelity, with timings from different code bases. At 2048×4096 SCA has no resolved ImageReward or FID advantage over AReA and trails on aesthetic score, CLIP-IQA, and patch FID/KID. The 191-prompt comparisons use one seed; extra seeds, SigMa, and progressive SCA cover 46 prompts, and FLUX.2-dev 50 prompts against full attention only. On FLUX.1-Krea-dev the ImageReward gain over AReA is unresolved and patch FID is higher. SigMa's dev-matched profile gives lattice-degraded images, possibly from our port or profile, not the method. We did not run ResDiT (no verified official code) or UltraImage's direct-extrapolation variant, so we claim no advantage over either. Late-appearing small extra subjects, 8192^2 grid artifacts, and one prompt's doubled limbs remain (Appendix Q), and the opening phase was tuned on development prompts. All metrics are automatic, low-sample FID is noisy, and intervals ignore multiple comparisons.

Conclusion. How much each head relies on a neighbourhood fixed in tokens versus a view of the whole image can be measured once at the training resolution and reused at 4K. With that table and a global-only opening, SCA is the strongest and fastest single-pass training-free operator we tested for FLUX.1-dev at 4096^2, at about a third of AReA's time. Multi-stage recipes lead on learned preference metrics through their schedules: as a drop-in operator in HRDiT's progressive schedule, SCA matches HRDiT's preference scores (no per-image metric significantly worse; HPSv2 and ARNIQA significantly better) in less than half its time.

References

Lorenzo Agnolucci, Leonardo Galteri, Marco Bertini, and Alberto Del Bimbo. ARNIQA: Learning distortion manifold for image quality assessment. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2024.

Black Forest Labs. FLUX.1. `https://github.com/black-forest-labs/flux`, 2024.

Black Forest Labs. FLUX.2. `https://huggingface.co/black-forest-labs/FLUX.2-dev`, 2025.

Black Forest Labs and Krea AI. FLUX.1 Krea [dev]. `https://huggingface.co/black-forest-labs/FLUX.1-Krea-dev`, 2025.

Jiazi Bu, Pengyang Ling, Yujie Zhou, Pan Zhang, Tong Wu, Xiaoyi Dong, Yuhang Zang, Yuhang Cao, et al. HiFlow: Training-free high-resolution image generation with flow-aligned guidance. *arXiv preprint arXiv:2504.06232*, 2025.

Juechu Dong, Boyuan Feng, Driss Guessous, Yanbo Liang, and Horace He. Flex attention: A programming model for generating optimized attention kernels. *arXiv preprint arXiv:2412.05496*, 2024.

Ruoyi Du, Dongliang Chang, Timothy Hospedales, Yi-Zhe Song, and Zhanyu Ma. DemoFusion: Democratising high-resolution image generation with no $$$. *arXiv preprint arXiv:2311.16973*, 2023.

Ruoyi Du, Dongyang Liu, Le Zhuo, Qin Qi, Hongsheng Li, Zhanyu Ma, and Peng Gao. I-Max: Maximize the resolution potential of pre-trained rectified flow transformers with projected flow. *arXiv preprint arXiv:2410.07536*, 2024.

Patrick Esser, Sumith Kulal, Andreas Blattmann, Rahim Entezari, Jonas Müller, Harry Saini, Yam Levi, Dominik Lorenz, et al. Scaling rectified flow transformers for high-resolution image synthesis. *arXiv preprint arXiv:2403.03206*, 2024.

Tianyu Fu, Haofeng Huang, Xuefei Ning, Genghan Zhang, Boju Chen, Tianqi Wu, Hongyi Wang, Zixiao Huang, et al. Mixture of attention spans: Optimizing LLM inference efficiency with heterogeneous sliding-window lengths. *arXiv preprint arXiv:2406.14909*, 2024.

Ali Hassani, Steven Walton, Jiachen Li, Shen Li, and Humphrey Shi. Neighborhood attention transformer. *arXiv preprint arXiv:2204.07143*, 2022.

Yingqing He, Shaoshu Yang, Haoxin Chen, Xiaodong Cun, Menghan Xia, Yong Zhang, Xintao Wang, Ran He, et al. ScaleCrafter: Tuning-free higher-resolution visual generation with diffusion models. *arXiv preprint arXiv:2310.07702*, 2023.

Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. GANs trained by a two time-scale update rule converge to a local Nash equilibrium. *arXiv preprint arXiv:1706.08500*, 2017.

Linjiang Huang, Rongyao Fang, Aiping Zhang, Guanglu Song, Si Liu, Yu Liu, and Hongsheng Li. FouriScale: A frequency perspective on training-free high-resolution image synthesis. *arXiv preprint arXiv:2403.12963*, 2024.

Noam Issachar, Guy Yariv, Sagie Benaim, Yossi Adi, Dani Lischinski, and Raanan Fattal. DyPE: Dynamic position extrapolation for ultra high resolution diffusion. *arXiv preprint arXiv:2510.20766*, 2025.

Zhiyu Jin, Xuli Shen, Bin Li, and Xiangyang Xue. Training-free diffusion model adaptation for variable-sized text-to-image synthesis. *arXiv preprint arXiv:2306.08645*, 2023.

Junjie Ke, Qifei Wang, Yilin Wang, Peyman Milanfar, and Feng Yang. MUSIQ: Multi-scale image quality transformer. *arXiv preprint arXiv:2108.05997*, 2021.

Yuval Kirstain, Adam Polyak, Uriel Singer, Shahbuland Matiana, Joe Penna, and Omer Levy. Pick-a-Pic: An open dataset of user preferences for text-to-image generation. *arXiv preprint arXiv:2305.01569*, 2023.

Sungho Koh, SeungJu Cha, Hyunwoo Oh, Kwanyoung Lee, and Dong-Jin Kim. ScaleDiff: Higher-resolution image synthesis via efficient and model-agnostic diffusion. *arXiv preprint arXiv:2510.25818*, 2025.

Yihua Liu, Fanjiang Ye, Bowen Lin, Rongyu Fang, and Chengming Zhang. TIDE: Text-informed dynamic extrapolation with step-aware temperature control for diffusion transformers. *arXiv preprint arXiv:2603.08928*, 2026.

Yiyang Ma, Feng Zhou, Xuedan Yin, Pu Cao, Yonghao Dang, and Jianqin Yin. ResDiT: Evoking the intrinsic resolution scalability in diffusion transformers. *arXiv preprint arXiv:2512.01426*, 2025. URL `https://arxiv.org/abs/2512.01426`.

Zizheng Pan, Jianfei Cai, and Bohan Zhuang. Fast vision transformers with HiLo attention. *arXiv preprint arXiv:2205.13213*, 2022.

William Peebles and Saining Xie. Scalable diffusion models with transformers. *arXiv preprint arXiv:2212.09748*, 2022.

Haonan Qiu, Shiwei Zhang, Yujie Wei, Ruihang Chu, Hangjie Yuan, Xiang Wang, Yingya Zhang, and Ziwei Liu. FreeScale: Unleashing the resolution of diffusion models via tuning-free scale fusion. *arXiv preprint arXiv:2412.09626*, 2024.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, et al. Learning transferable visual models from natural language supervision. *arXiv preprint arXiv:2103.00020*, 2021.

Javad Rajabi, Koorosh Roohi, David B. Lindell, and Babak Taati. AReA: Adaptive receptive-field attention for efficient high-resolution diffusion transformers. *arXiv preprint arXiv:2610.17227*, 2026a. URL https://arxiv.org/abs/2610.17227.

Javad Rajabi, Kimia Shaban, Koorosh Roohi, David B. Lindell, and Babak Taati. SEGA: Spectral-energy guided attention for resolution extrapolation in diffusion transformers. *arXiv preprint arXiv:2605.22668*, 2026b.

Tim Salimans, Ian Goodfellow, Wojciech Zaremba, Vicki Cheung, Alec Radford, and Xi Chen. Improved techniques for training GANs. In *Advances in Neural Information Processing Systems*, 2016.

Christoph Schuhmann. LAION aesthetics predictor. https://github.com/LAION-AI/aesthetic-predictor, 2022.

Ethan Smith, Nayan Saxena, and Aninda Saha. ToDo: Token downsampling for efficient generation of high-resolution images. *arXiv preprint arXiv:2402.13573*, 2024.

Jianlin Su, Yu Lu, Shengfeng Pan, Ahmed Murtadha, Bo Wen, and Yunfeng Liu. RoFormer: Enhanced transformer with rotary position embedding. *arXiv preprint arXiv:2104.09864*, 2021.

Jianyi Wang, Kelvin C. K. Chan, and Chen Change Loy. Exploring CLIP for assessing the look and feel of images. *arXiv preprint arXiv:2207.12396*, 2022.

Xiaoshi Wu, Yiming Hao, Keqiang Sun, Yixiong Chen, Feng Zhu, Rui Zhao, and Hongsheng Li. Human preference score v2: A solid benchmark for evaluating human preferences of text-to-image synthesis. *arXiv preprint arXiv:2306.09341*, 2023.

Haocheng Xi, Shuo Yang, Yilong Zhao, Chenfeng Xu, Muyang Li, Xiuyu Li, Yujun Lin, Han Cai, et al. Sparse VideoGen: Accelerating video diffusion transformers with spatial-temporal sparsity. *arXiv preprint arXiv:2502.01776*, 2025.

Guangxuan Xiao, Jiaming Tang, Jingwei Zuo, Junxian Guo, Shang Yang, Haotian Tang, Yao Fu, and Song Han. DuoAttention: Efficient long-context LLM inference with retrieval and streaming heads. *arXiv preprint arXiv:2410.10819*, 2024.

Jiazheng Xu, Xiao Liu, Yuchen Wu, Yuxuan Tong, Qinkai Li, Ming Ding, Jie Tang, and Yuxiao Dong. ImageReward: Learning and evaluating human preferences for text-to-image generation. *arXiv preprint arXiv:2304.05977*, 2023.

Yu Xue, Haoxuan Qu, Zhuoling Li, Hongbin Xu, Jianxiong Yin, Simon See, Hossein Rahmani, and Jun Liu. HRDiT: Training-free high-resolution image generation with off-the-shelf diffusion transformer models. *arXiv preprint arXiv:2608.07003*, 2026. URL https://arxiv.org/abs/2608.07003.

Haosen Yang, Adrian Bulat, Isma Hadji, Hai X. Pham, Xiatian Zhu, Georgios Tzimiropoulos, and Brais Martinez. FAM diffusion: Frequency and attention modulation for high-resolution image generation with stable diffusion. *arXiv preprint arXiv:2411.18552*, 2024.

Zhihang Yuan, Hanling Zhang, Pu Lu, Xuefei Ning, Linfeng Zhang, Tianchen Zhao, Shengen Yan, Guohao Dai, et al. DiTFastAttn: Attention compression for diffusion transformer models. *arXiv preprint arXiv:2406.08552*, 2024.

Jinjin Zhang, Qiuyu Huang, Junjie Liu, Xiefan Guo, and Di Huang. Diffusion-4K: Ultra-high-resolution image synthesis with latent diffusion models. *arXiv preprint arXiv:2503.18352*, 2025a.

Peiyuan Zhang, Yongqi Chen, Runlong Su, Hangliang Ding, Ion Stoica, Zhengzhong Liu, and Hao Zhang. Fast video generation with sliding tile attention. *arXiv preprint arXiv:2502.04507*, 2025b.

Shen Zhang, Zhaowei Chen, Zhenyu Zhao, Yuhao Chen, Yao Tang, and Jiajun Liang. HiDiffusion: Unlocking higher-resolution creativity and efficiency in pretrained diffusion models. *arXiv preprint arXiv:2311.17528*, 2023.

Bingxuan Zhao, Qing Zhou, Yu Wang, Chuang Yang, and Qi Wang. σ: Sigmoid modulation for ultra high resolution diffusion. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pp. 162993–163008. PMLR, 2026. URL https://proceedings.mlr.press/v306/zhao26cc.html.

Min Zhao, Bokai Yan, Xue Yang, Hongzhou Zhu, Jintao Zhang, Shilong Liu, Chongxuan Li, and Jun Zhu. UltraImage: Rethinking resolution extrapolation in image diffusion transformers. *arXiv preprint arXiv:2512.04504*, 2025. URL https://arxiv.org/abs/2512.04504.

Le Zhuo, Ruoyi Du, Han Xiao, Yangguang Li, Dongyang Liu, Rongjie Huang, Wenze Liu, Lirui Zhao, et al. Lumina-Next: Making Lumina-T2X stronger and faster with Next-DiT. *arXiv preprint arXiv:2406.18583*, 2024.

APPENDIX CONTENTS

Method
A Proofs 13
B Calibration details and statistics 14
C Implementation details and cost 16
D Hyperparameters 18
Evaluation setup
E Evaluation protocol and metrics 19
F Reproducing AReA 20
G Other baselines and FLUX.2-dev 21
Quantitative results
H Extended results with intervals 22
I Additional paired intervals 27
J Sampling-budget, guidance, and schedule checks 30
K Exploratory 8192^2 comparison 33
L Calibration table at non-square sizes 34
M The low-band duplication guard 34
Compute and impact
N Compute and reproducibility 35
O Broader impact 35
Qualitative results
P Qualitative galleries 36
Q Failure cases 47

A PROOFS

Lemma 1 (Softmax split)**.** *Let the keys of a query be partitioned into a near set $\mathcal{K}^{\text{near}}$ and a far set $\mathcal{K}^{\text{far}}$, with logits ℓ_j. Then $O^{\text{full}} = m\,O^{\text{near}} + (1-m)\,O^{\text{far}}$ with $m = \sum_{j\in\mathcal{K}^{\text{near}}} e^{\ell_j} / \sum_j e^{\ell_j}$ and $O^{\text{near}}, O^{\text{far}}$ the softmax outputs restricted to $\mathcal{K}^{\text{near}}$ and $\mathcal{K}^{\text{far}}$.*

Proof. Write $Z^{\text{near}} = \sum_{j\in\mathcal{K}^{\text{near}}} e^{\ell_j}$, so that $Z^{\text{near}}O^{\text{near}} = \sum_{j\in\mathcal{K}^{\text{near}}} e^{\ell_j} v_j$, and likewise for the far set. Then $O^{\text{full}} = (Z^{\text{near}}O^{\text{near}} + Z^{\text{far}}O^{\text{far}})/(Z^{\text{near}} + Z^{\text{far}})$, and $Z^{\text{near}}/(Z^{\text{near}} + Z^{\text{far}}) = m$. □

Proof of Proposition 1. Global operator. Image queries sit at $c(p_i) \in \mathcal{G}_0 = [n_0]^2$ and text queries at $0 \in \mathcal{G}_0$; subsampled keys sit at $b \in \mathcal{G}_0$ and text keys at 0. Any two points of $[n_0]^2$ differ by at most $n_0 - 1$ per axis. The key count is $T + |\mathcal{G}_0| = T + n_0^2$.

Local operator. Fix one axis of length $H \geq w$ and a tile $[t\tau,\, t\tau + t)$ with $t \leq w$. The unclamped start $a' = t\tau + \lfloor t/2 \rfloor - \lfloor w/2 \rfloor$ satisfies $a' \leq t\tau$ because $\lfloor t/2 \rfloor \leq \lfloor w/2 \rfloor$, and $a' + w \geq t\tau + t$ because $w - \lfloor w/2 \rfloor = \lceil w/2 \rceil \geq \lceil t/2 \rceil = t - \lfloor t/2 \rfloor$. If $a' < 0$ the window is $[0, w)$, and $w \geq a' + w \geq t\tau + t$. If $a' > H - w$ the window is $[H - w, H)$, with $H - w < a' \leq t\tau$ and $H \geq t\tau + t$. In every case the window contains the tile, so it contains each query of the tile, and two positions in a window of side $w \leq n_0$ differ by at most $n_0 - 1$. The key count is $T + w^2 = T + n_0^2$ when $n_0 \leq \min(H, W)$. □

Remark 1 (Text-image offsets). FLUX.1 places all text tokens at position 0, so the offset between an image query at p_i and a text key is p_i. In the global operator the query sits at $c(p_i) \in [n_0]^2$, inside the trained range. In the local operator it keeps p_i, up to $sn_0 - 1$ per axis, exactly as in full attention and AReA's windows at the target. Mapping text offsets through $c(\cdot)$ in the local operator is a possible refinement that we have not tested.

Proof of Proposition 2. Full attention scores every query against every key, $(N + T)^2$ pairs. The global operator scores $N + T$ queries against $T + |\mathcal{G}_0|$ keys with $|\mathcal{G}_0| = N/(s_H s_W) = n_0^2$. The local operator scores N image queries against T text keys and w^2 window keys. The sum is at most $(2N + T)(T + w^2 + N_0)$, which is $O(N(T + w^2 + N_0))$ for fixed n_0, T. Value aggregation has the same pair counts. In the opening phase the local term is absent. The FlexAttention kernel visits whole blocks and masks the extra pairs: a query block of 8×16 tokens spans two tiles whose windows are 8 columns apart, so it visits a 64×72 key region, 12.5% more than one window, plus partial blocks at clamped borders. This changes the constant, not the order. □

Proof of Proposition 3. $J(\pi) = E - 2\pi B + \pi^2 A$ with $A \geq 0$. If $A = 0$ then $O^{\text{loc}} = O^{\text{glob}}$ on every summand, so $B = 0$ and J is constant; any π, in particular $\pi^\star = 1$, is optimal and the bound holds trivially. If $A > 0$, J is a strictly convex quadratic with unconstrained minimizer B/A, and its minimizer on $[0, 1]$ is the projection $\text{clip}(B/A, 0, 1)$. For any π',

$$J(\pi') - J(\pi^\star) = A(\pi' - \pi^\star)^2 + 2(\pi' - \pi^\star)(A\pi^\star - B).$$

If $0 < \pi^\star < 1$ then $A\pi^\star = B$ and the second term vanishes. If $\pi^\star = 0$ then $B \le 0$, so $A\pi^\star - B \ge 0$ and $\pi' - \pi^\star \ge 0$. If $\pi^\star = 1$ then $B \ge A$, so $A\pi^\star - B \le 0$ and $\pi' - \pi^\star \le 0$. In both clipped cases the second term is nonnegative.

Any rule that assigns one weight per cell, such as a per-layer table, a weight shared per noise level, a hard switch, or a table fitted elsewhere, is a feasible π' in every cell, so its total error is at least the per-head table's, by $\sum_{\text{cells}} A(\pi' - \pi^\star)^2$ or more. A table interpolated in σ between the mean sampled noise levels of occupied bins is not constant within a bin; the statement applies to the bin-constant table used to score rules in Section 3. □

The frequency argument in more detail. Argument 1 rests on two facts and one assumption. The first fact is that the noisy latent is a sum of signal and independent white noise, so a band's signal-to-noise ratio scales as $((1-\sigma)/\sigma)^2 S(f)$; at $\sigma = 0.89$ the factor is 0.015, and at $\sigma = 0.5$ it is 1. The second is that a window of w tokens cannot distinguish a component of period longer than w from a smooth trend within the window, so windows cannot coordinate a layout at that scale except through what earlier layers mixed into their keys. The assumption is that natural-image latent spectra fall with frequency, which we have not measured for FLUX.1 latents. Together they predict that windowed attention fails first and worst at high noise and for objects larger than the window, which matches the duplication we observe (Figure 4) and the effect of a 16-token window.

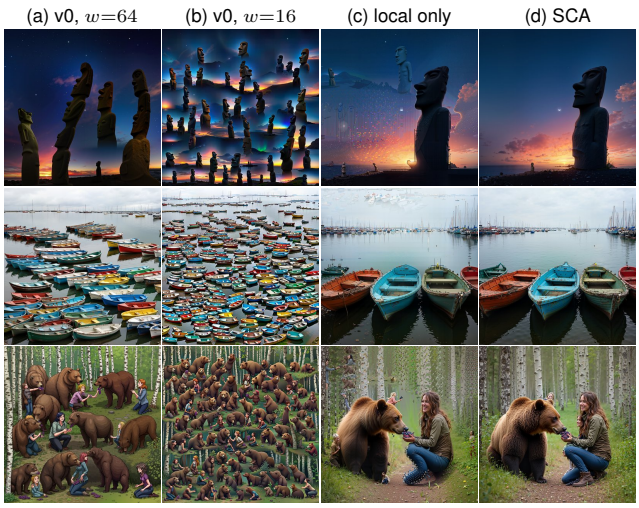

Figure 4: **Windows at high noise duplicate subjects.** 4096^2, three development prompts. (a) v0, the calibrated mix from the first step: each 64-token window settles the layout on its own. (b) A 16-token window gives smaller, more numerous copies. (c) With the opening phase but local-only attention afterwards ($\pi = 1$), ghost copies and grid speckle remain. (d) SCA.

B Calibration details and statistics

Statistics. For every emulated scale, block, head, and noise bin, the pass stores A, B, E, the error of the local branch alone $\sum \|O^{\text{full}} - O^{\text{loc}}\|^2$, $\sum \|O^{\text{full}}\|^2$, and the query count, in float64, summed over image queries and channels (`sca/calib.py`). The sampler's trajectory is that of unmodified full attention; the emulated operators only observe. Statistics are written per prompt and merged by summation, so prompts can be added later and split-half fits need no extra pass. Text-query outputs are not in the objective, because SCA sends text queries through the global branch only.

Geometry at emulated scale. At 1024^2 the grid is 64×64. Emulated scale 2 uses a 32×32 window, 2×2 subsampling onto a 32×32 grid, and 4×4 query tiles; emulated scale 4 uses a 16×16 window, 4×4 subsampling onto a 16×16 grid, and 2×2 tiles. At 1280^2 the grid is 80×80, and emulated scale 2 uses a 40×40 window, 2×2 subsampling onto a 40×40 grid, and 5×5 tiles.

Noise bins. With 28 steps and $\mu = 1.15$, σ runs from 1.0 to 0.105, so the lowest of the 14 bins is empty and copies its neighbour; 13 bins are populated. Under this schedule 8 steps have $\sigma > 0.89$ (the ninth has $\sigma = 0.888$) and 13 have $\sigma > 0.80$.

No global phase in output energy. The best single shared weight falls from 0.88 at $\sigma = 0.10$ to 0.72 at $\sigma = 0.97$, and the mean per-head weight from 0.74 to 0.64 (Figure 7); within a block, the

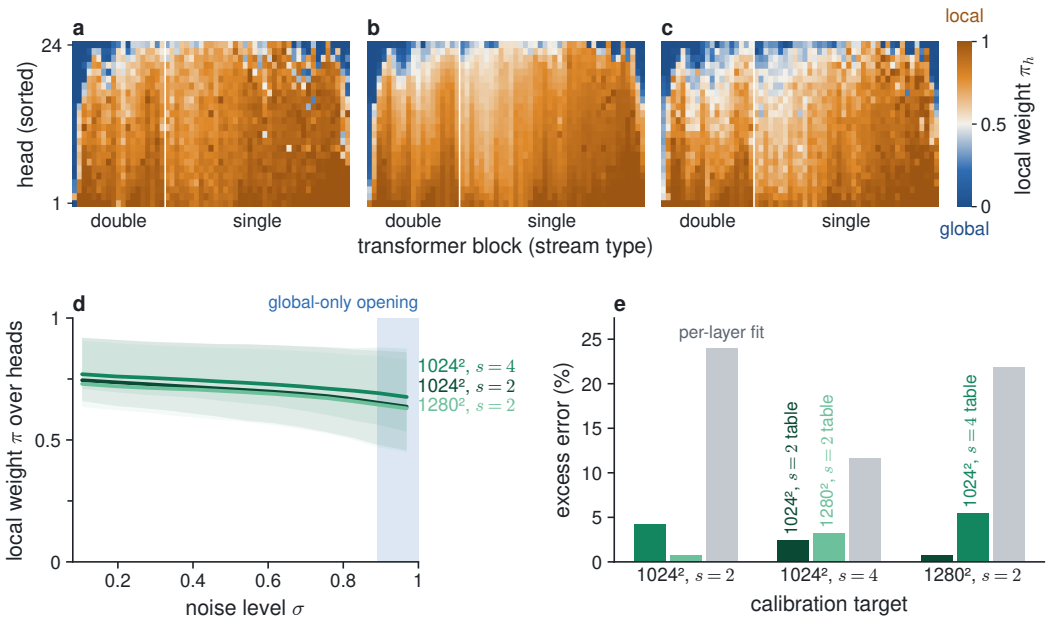

Figure 5: **Heads differ widely, yet a calibrated table transfers across scale and resolution.** (a to c) $\pi_h(\sigma)$ at $\sigma = 0.10, 0.46, 0.83$ ($\sigma = 1$: pure noise); (d) mean and interquartile range across heads, shaded: global-only opening. (e) Excess error over each target's table: at most 5.4% transferred, 12 to 24% per-layer.

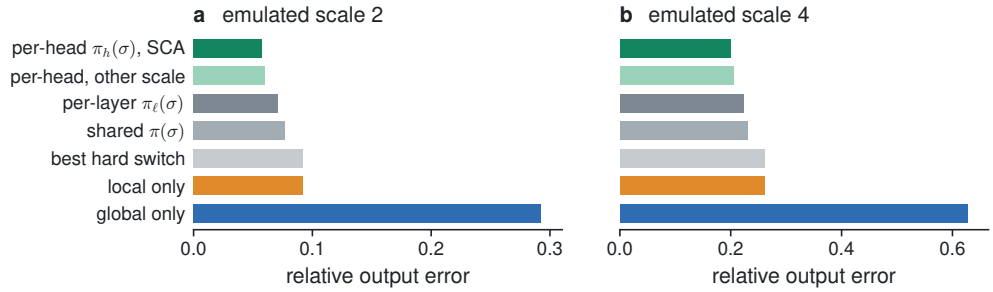

Figure 6: **Per-head weights explain full attention best, and transfer across scales.** Relative output error per mixing rule at emulated scales 2 (a) and 4 (b), 32 prompts; "other scale" swaps the scale-2 and scale-4 tables; the best hard switch ($\sigma^\star = 1$) equals local only. In-sample and held-out errors agree within 0.0003.

per-head weights span a median range of 0.66 to 0.84 at every noise level (Figure 8). Among hard switches from global to local, the best threshold is $\sigma^\star = 1$, which equals local only. A switch at $\sigma^\star = 0.89$ between our two branches raises the error from 0.092 to 0.152 at scale 2 and from 0.261 to 0.376 at scale 4. In the noisiest bin, the global branch alone has relative error 0.27 at scale 2, the local branch alone 0.14, and the per-head mix 0.09. The generation results of Section 4.2 show the opposite preference for the opening steps, which is the gap between squared output error and layout.

Attention mass. For 256 random image queries per block and step, the pass also records the softmax mass on text keys and on image keys within Chebyshev distance 2, 4, 8, and 16. At 1024^2, averaged over heads and queries, text receives 15 to 19% of the mass; image keys within distance 16 receive 52% at $\sigma = 0.10$ and 40% at $\sigma = 0.97$, and keys outside that 33×33 neighbourhood still receive 29% and 41%. A purely local receptive field cannot account for any step, even at native resolution.

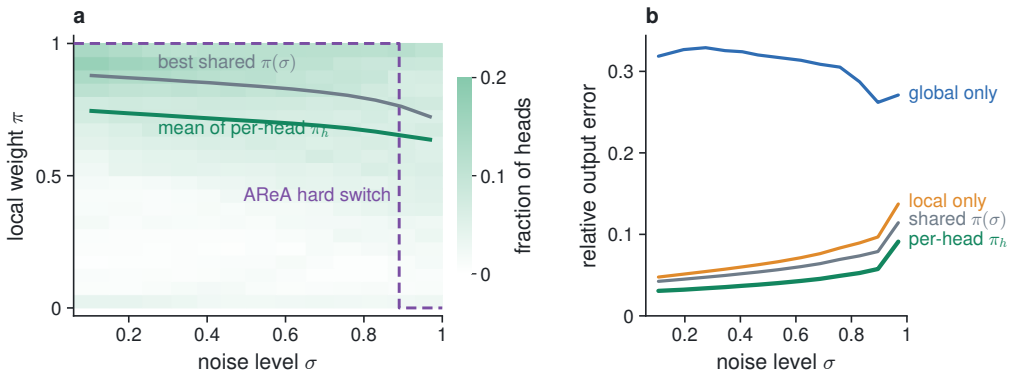

Figure 7: **In output energy, no noise level prefers the global branch.** (a) Fraction of heads per π and noise bin, with mean $\pi_h(\sigma)$, best shared $\pi(\sigma)$ and AReA's switch; scale 2, 32 prompts, $\sigma = 1$: pure noise. (b) Error per noise bin: local beats global everywhere; per-head mixing beats both.

Calibration budget. We merge the stored sufficient statistics from the first 4, 8, or 16 calibration prompts and rerun the 46 evaluation prompts with each resulting table. Table 19 reports generation-level differences from the 32-prompt table. The test reuses calibration observations but uses separate prompts for generation evaluation; it is not a held-out output-error benchmark.

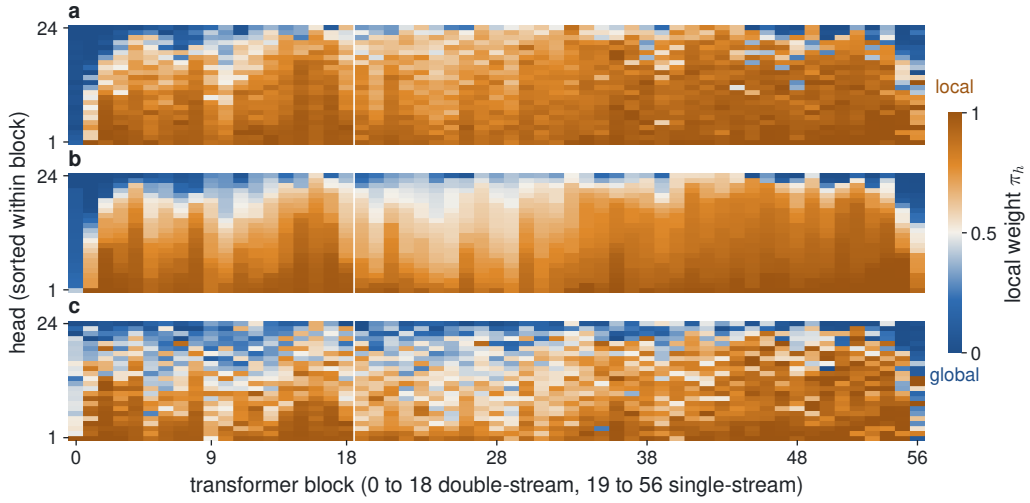

Figure 8: **Heads in the same block differ widely at every noise level.** $\pi_h(\sigma)$ of the scale-2 table at $\sigma = 0.10$ (a), 0.54 (b), 0.97 (c), 57 blocks by 24 heads, 32 prompts, heads sorted per block by (b). Median within-block range 0.66 to 0.84; Figure 5 shows the scale-4 inference table.

C Implementation details and cost

FlexAttention window. Image tokens are reordered tile-major, so that a sparse block is a small spatial tile. Query blocks cover 8×16 tokens (128 queries) and key blocks 8×8 (64 keys), shifted so that unclamped windows start on a block edge; text keys come first. The mask function states window membership per query-key pair, so the result does not depend on the block layout, only the amount of skipped work does. On GH200, 8×16 query blocks with 8×8 key blocks were fastest at 2048^2, 4096^2, and 2048×6144. RoPE, subsampling, both branches, and the mix compile into one graph per layer; the mix is in fp32.

Algorithm 1 SCA: calibration (once per model) and one attention layer at sampling time.

Calibrate(prompts, s_c) on an in-distribution $n \times n$ grid:
sample with full attention; at every block l, step σ, and head h:
  $O^{\text{glob}}, O^{\text{loc}} \leftarrow$ Eqs. 2, 3 with subsampling s_c, window n/s_c, tile $tn/(s_c n_0)$
  add $\|O^{\text{loc}} - O^{\text{glob}}\|^2$ and $\langle O^{\text{full}} - O^{\text{glob}}, O^{\text{loc}} - O^{\text{glob}} \rangle$ to $A_{lh\beta(\sigma)}$, $B_{lh\beta(\sigma)}$ ▷ float64
return $\pi_{lh\beta} \leftarrow \text{clip}(B_{lh\beta}/A_{lh\beta}, 0, 1)$ ▷ Proposition 3
Layer(Q, K, V, σ) at target factors (s_H, s_W):
$O^{\text{glob}} \leftarrow$ Eq. 2 for all queries ▷ block-centre keys, coarse positions
if $\sigma > \sigma_g$ **then return** O^{glob} ▷ opening phase
$O^{\text{loc}} \leftarrow$ Eq. 3 for image queries ▷ one FlexAttention call
return $[\, O^{\text{glob}}_{\mathcal{T}},\ \pi_h(\sigma)\, O^{\text{loc}} + (1 - \pi_h(\sigma))\, O^{\text{glob}}_{\mathcal{G}} \,]$ ▷ interpolated table; fp32 mix

Numerical checks. The 18 CPU tests in `tests/test_sca.py` use float32 to check full-mode agreement with stock FLUX attention, both branches against brute-force Equations 2 and 3, full-grid and factor-1 identities, calibration when the local branch is exact, the gate, and FlexAttention against an explicit gather reference. A separate GPU check in `scripts/bench_attn.py` compares the gather and FlexAttention implementations in bf16: the recorded maximum absolute difference is 0.015625, and both have the same maximum error against the fp32 reference on the tested grids. This is an observed absolute-error check; no per-element ULP bound was measured.

Position ids in bf16. The FLUX.1 pipeline builds image position ids in bf16, which cannot represent every integer above 256. Our frozen runs retain these rounded ids for RoPE. Passing the true grid from the sampler fixes window membership and subsampling geometry, but does not restore exact RoPE coordinates: the coarse-query positions are also computed from the rounded ids. At a 384-token axis, index 383 rounds to 384 and its scale-6 coarse coordinate becomes 64; at a 512-token axis, index 511 rounds to 512 and its scale-8 coarse coordinate also becomes 64. A local 64-token window can likewise have a rounded endpoint difference of 64 instead of the ideal bound 63. Proposition 1 therefore applies exactly to the mathematical operator, not to these finite-precision positions at 2048×6144, 6144^2, or 8192^2. The 4096^2 grid uses indices 0–255, which bf16 represents exactly. FLUX.2 uses integer position ids. Rebuilding FLUX.1 positions in higher precision would require a separate controlled experiment; the reported runs are unchanged.

Non-divisible grids. When H/n_0 is not an integer, the subsampling factor per axis is the divisor of H nearest to H/n_0, and the tile side the divisor nearest to t; neither case arises at the sizes we evaluate.

Table 4: **Cost.** Top: analytic attention FLOPs per image (57 blocks, 24 heads, head dimension 128, 28 steps, $T = 512$); "gated" skips the local branch in the first 8 steps; ratio is full over gated SCA. Bottom: measured time of one attention layer in bf16 on GH200, minimum over interleaved runs; SCA is both branches and the mix.

Size ($H \times W$)	Grid	(s_H, s_W)	Full	SCA, no gate	SCA, gated	Ratio
2048×4096	128×256	(2, 4)	2.17×10^{16}	5.97×10^{15}	5.12×10^{15}	4.2
4096×4096	256×256	(4, 4)	8.56×10^{16}	1.19×10^{16}	1.02×10^{16}	8.4
2048×6144	128×384	(2, 6)	4.84×10^{16}	8.93×10^{15}	7.66×10^{15}	6.3

Token grid	Pixels	Full (SDPA, ms)	SCA (ms)	Speedup
128×128	2048^2	7.1	5.2	1.4
256×256	4096^2	87.3	18.7	4.7
128×384	2048×6144	49.6	14.0	3.5

End-to-end latency. Table 5 comes from one matched protocol on one GH200 (`results/latency_curve_batch1.csv`, which also lists the excluded warm-up images; `results/latency_curve_summary.csv` summarizes all cells but full attention at 6144^2). At each size, each method generates one excluded warm-up image and then three timed images in the same process, at batch 1 and 28 steps; we report the median. At 4096^2 the medians are 186.5 s for full

Table 5: **Seconds per image**, batch 1, idle GH200, 28 steps with text encoding and VAE decoding, median of three timed images after an excluded warm-up; full attention not run at 8192^2.

Size (N tokens)	Full attention	AReA	SCA
2048^2 (16,384)	22.9	41.0	**18.2**
3072^2 (36,864)	73.9	114.3	**40.2**
4096^2 (65,536)	186.5	202.4	**71.1**
6144^2 (147,456)	762.7	482.3	**161.6**
8192^2 (262,144)	not run	952.0	**287.6**

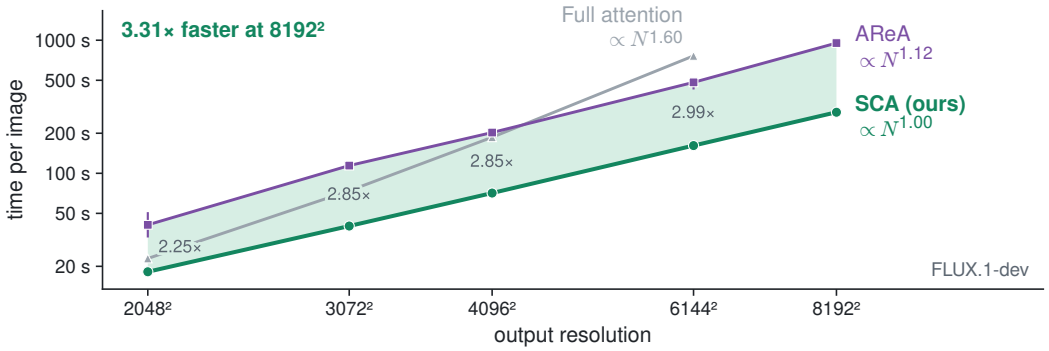

Figure 9: **SCA is fastest at every size; its lead over AReA grows to 3.31× at 8192^2.** Batch 1 on one GH200, median of three prompts, whiskers min to max; full attention was not run at 8192^2. Grey ratios: speedup over AReA per size; exponents fit seconds $\propto N^b$ (N image tokens).

attention (186.0, 186.6, 186.5), 202.4 s for AReA (192.7, 202.4, 204.3), and 71.1 s for SCA (71.0, 71.1, 71.1). At 2048^2 and 3072^2 they are 22.9 and 73.9 s for full attention, 41.0 and 114.3 s for AReA, and 18.2 and 40.2 s for SCA. At 6144^2 and 8192^2, AReA takes 482.3 and 952.0 s and SCA 161.6 and 287.6 s, so SCA reduces measured time by 66.5% and 69.8%. At 6144^2 full attention takes 762.7 s (763.5, 762.7, 762.5), 4.7× SCA's time. An earlier 4096^2 benchmark, kept as a separate repeat (`results/latency_4096_batch1.csv`), gave 184.6 s for full attention (184.1, 184.7, 184.6), 212.6 s for AReA (207.3, 225.8, 212.6), 88.1 s for SCA + band guard (88.1, 98.7, 88.1; single-pass compiled band), and 71.0 s for SCA final (71.0, 83.1, 70.9). On the shared campaign nodes, per-image times at 4096^2 were 71 s for SCA final, 120 s for SCA + band guard, and 262 s for AReA; the campaign code used an unfused band, so dropping it saved more there (49 s) than in the clean measurement (17 s). At the non-square sizes we have only campaign timings, with both methods on the same pool of shared nodes (mean over 191 prompts): SCA takes 38.1 s and AReA 112.4 s at 2048×4096, and 55.8 s and 150.5 s at 2048×6144.

D HYPERPARAMETERS

Table 6 lists every setting of the main runs.

Table 6: **All hyperparameters** of the main runs (code commit `2d11740`), produced by `scripts/sample.py --method sca --pi-table calib/k1_1024.npz --gate-sigma 0.89`. Appendix G gives the settings that differ for UltraImage, HRDiT, SigMa, and FLUX.2-dev.

Group	Setting	Value
Model	checkpoint, precision	FLUX.1-dev, bf16 weights and activations
	blocks, heads, head dim	19 double-stream + 38 single-stream, 24, 128
	native grid n_0, text tokens T	64 (1024^2 pixels, 16 pixels per token), 512
Sampler	steps, solver	28, flow-matching Euler
	guidance, shift	4.5 (distilled guidance), $\mu = 1.15$ at every size
	seed, decoding	seed 0 per prompt, shared across methods; tiled VAE decoding
Local branch	tile t, window w	8, 64 (clamped at borders), original positions
	kernel	FlexAttention; query blocks 8×16, key blocks 8×8, tile-major order
Global branch	subsampling	block centre, factor (s_H, s_W) = target grid / n_0
	positions	queries at $c(p)$, keys at block index, text at 0
Mixing	table	per (block, head, σ bin), $57 \times 24 \times 14$, scale-4 entry
	interpolation	linear in σ between the mean sampled noise levels of occupied bins, constant outside
	gate	$\sigma_g = 0.89$ (8 of 28 steps global only)
Calibration	prompts, resolution	32 at 1024^2 (16 more at 1280^2 for transfer)
	emulated scales	2 and 4, one pass; 14 bins of width $1/14$
	hardware	one GH200; complete calibration runtime was not retained
Band guard	(not in final method)	average-pool $p = 16$, bilinear upsampling

E Evaluation protocol and metrics

Table 7: **Evaluation protocol.** The FLUX.1-dev attention comparisons share sampler settings and prompt seeds. UltraImage and HRDiT use their own multi-stage protocols; FLUX.2-dev uses its own native-resolution schedule (Appendix G). Prompt counts exclude four development prompts. Table 6 lists the hyperparameters; the paragraphs below define the metrics.

Models	FLUX.1-dev (Black Forest Labs, 2024) (main results); FLUX.1-Krea-dev (Black Forest Labs & Krea AI, 2025) (100 prompts); FLUX.2-dev (Black Forest Labs, 2025) (50 prompts); bf16.
Sampler	28 Euler flow-matching steps, guidance 4.5, shift $\mu = 1.15$ at every size, seed label 0 per prompt, tiled VAE decoding; UltraImage, HRDiT, and FLUX.2-dev exceptions are detailed in Appendix G.
Sizes	4096^2, 2048×4096, 2048×6144 (191 prompts each); 6144^2 (100 prompts); 8192^2 (12 exploratory paired prompts, metrics and qualitative examples).
Prompts	Aesthetic-4K evaluation set with 4K references (Zhang et al., 2025a), 191 prompts, fixed 46-prompt ablation subset; HRGen-1K (Rajabi et al., 2026a), 200 prompts.
Baselines	Full attention; AReA, released code (Rajabi et al., 2026a); DyPE (Issachar et al., 2025), SEGA (Rajabi et al., 2026b), UltraImage (two-stage defaults; Zhao et al., 2025); on the 46-prompt subset also SigMa (Zhao et al., 2026) and HRDiT (released progressive protocol; Xue et al., 2026).
Metrics	ImageReward, HPSv2/2.1, PickScore, CLIP, aesthetic; FID, KID, IS, patch FID/KID; MUSIQ (full, 1024^2 crops), CLIP-IQA, ARNIQA; 95% paired bootstrap intervals over prompts.
Hardware	One NVIDIA GH200 per process; verified timing logs identify the GH200 120 GB model. Clean latency uses batch 1; campaign timings are marked separately.

Prompts and references. The Aesthetic-4K evaluation split (Zhang et al., 2025a) has 195 prompts with 4K reference images. Four (boats, Moai, woman and bear, underwater surfer) were used to choose the opening phase and window and are excluded from generation evaluation, leaving 191. Distribution metrics retain all 195 official reference rows. The 46-prompt ablation subset was fixed

before any ablation was run. HRGen-1K (Rajabi et al., 2026a) has 1000 prompts and no references; we use source rows 0–199, with ids `hrgen_0000`–`hrgen_0199`. The 48 calibration prompts were written for this purpose and share no prompt with Aesthetic-4K (the largest word-set Jaccard overlap with any evaluation prompt is 0.26).

Preference and alignment metrics. ImageReward v1.0 (Xu et al., 2023) (bicubic resize to 224, centre crop). HPSv2 and HPSv2.1 (Wu et al., 2023): cosine of normalized image and text features with the official validation transform (longest side 224, padded). PickScore (Kirstain et al., 2023): $\exp(\text{logit scale})$ times the cosine. CLIP score (Radford et al., 2021): $100\max(\cos, 0)$ with ViT-L/14. Aesthetic: the LAION improved aesthetic predictor (sac+logos+ava1, linear MSE) on the L2-normalized CLIP ViT-L/14 image embedding (Schuhmann, 2022).

No-reference quality. MUSIQ (Ke et al., 2021) (pyiqa defaults) on the full image at native resolution, and MUSIQ_c, its mean over four fixed random 1024^2 crops per image; crop positions depend only on the prompt id, so they are identical across methods. CLIP-IQA (Wang et al., 2022) and ARNIQA (Agnolucci et al., 2024) (pyiqa defaults) on the image downscaled with antialiased bicubic resampling to a long side of 1024 pixels, the scale both were trained on.

Distribution metrics and reference correction. FID (Heusel et al., 2017) and KID compare the generated images (191, or 46 on the subset) with all 195 official reference rows, using Inception-v3 features (torch-fidelity) after bilinear resizing to 299^2. The reference rows contain 168 distinct image contents; repeated rows retain their official weights. Patch metrics use four 1024^2 crops per row (764 generated and 780 reference crops). An audit found that the original feature-bank assembly keyed reference crops by resolved image path, so repeated images reused the last row's crop features. We corrected the lookup to retain each row's intended cached crops. Whole-image FID/KID are unchanged; all patch point estimates reported here use the corrected assembly. Original CSVs remain archived for provenance. Inception score uses full images at 299^2, with 10 splits.

Intervals. Per-image intervals are 95% paired percentile bootstraps over prompt ids: original summaries use 1000 draws; separately recomputed comparisons use 2000 draws with seed 0. For the main SCA–AReA FID/KID and patch-FID/KID comparison, 1000 draws resample generated prompt ids jointly across methods and resample the 168 distinct reference-image identities, keeping all rows and crops from each sampled identity together. This retains the 195-row weighted point estimate while accounting for repeated reference images. The resulting FID difference is -7.63 $[-12.79, -2.94]$; patch FID is -3.60 $[-7.39, 1.15]$ and patch KID ($\times 10^3$) is -2.27 $[-4.73, 0.12]$. An earlier version of this analysis reported a significant patch-KID improvement; under the corrected evaluation its interval includes zero. The whole-image and patch FID/KID comparisons at both non-square sizes and on the 46-prompt panel, the added baselines' patch comparisons, and the 56-step patch-FID comparison also use reference-image clusters. Other whole-image FID/KID intervals are labelled historical reference-row bootstraps; they have not been recomputed with image-identity clusters. Historical patch intervals are not attached to corrected estimates and are omitted where recomputation is incomplete. FID's finite-sample bias can place point estimates outside percentile intervals. Intervals are unadjusted for multiple comparisons and measure prompt/reference sampling uncertainty, not variability over independent model training runs.

F Reproducing AReA

We run AReA's released code with its FLUX.1 defaults (block-centre pooling, first phase for 8 of 28 steps, halo width 8, bridge ratio 0.5), the FLUX.1-dev checkpoint, guidance 4.5, 28 steps, μ at its maximum, and the seeds shared with the other methods. AReA's paper reports ImageReward 1.32 at 4K under its own settings; our shared protocol gives 0.92, so cross-paper numbers are not comparable, while our direct SCA–AReA comparison holds these settings fixed. UltraImage and FLUX.2-dev use separate protocols (Appendix G). AReA's paper reports 151 s for AReA and 314 s for full attention at 4096^2 on an H100; the matched sweep measures 202.4 s and 186.5 s on one GH200 (medians of three images). We use commit `9e38ec2` of the released repository, which includes a cost cap on halo and bridge tokens and a jagged halo edge; neither appears in AReA's paper, which does not say whether its Table 1 used them. Recomputed from the code with 512 text tokens, the first-phase pooling factors are 4, 4, 4, 4, 2, 2, 2, 2 at 4096×4096 and 2048×4096,

and 8, 8, 8, 4, 4, 4, 2, 2 at 2048×6144. The second-phase windows shrink over 20 steps from 180, 151, and 192 tokens respectively to 64. At 2048×6144 no window is cheaper than full attention once the text tokens are counted, so the first window falls back to 192 tokens and the cap drops the bridge and halo on the first three second-phase steps. Other differences from the release defaults are the FLUX.1-dev checkpoint instead of FLUX.1-Krea-dev, and one seed per prompt shared across methods instead of one seed for all prompts. On FLUX.1-Krea-dev, AReA's default checkpoint, we rerun AReA and recalibrate SCA (16 prompts at 1024^2, scale-4 table, other settings unchanged) with the same 28-step, guidance-4.5 sampler on 100 prompts; DyPE and SEGA run the Krea transformer through the frozen baseline harness. There SCA matches AReA at a third of its time (Table 16).

G Other baselines and FLUX.2-dev

DyPE and SEGA. We use the existing baseline harness frozen at commit `c05ebdc`, with FLUX.1-dev, 28 steps, guidance 4.5, maximum shift, tiled VAE decoding, and the same prompt-derived seeds as SCA. These settings isolate the attention/position treatment on the checkpoint used in our main comparison. They do not reproduce the methods' reported FLUX.1-Krea-dev results; on Krea weights both trail AReA on ImageReward (Table 16). Their low FLUX.1-dev scores should therefore not be read as a ranking under each method's preferred checkpoint.

UltraImage. We run a guided two-stage configuration from the FLUX branch of the released implementation through `scripts/run_ultraimage.py`. It uses a 1024^2 first stage and a 4096^2 second stage, 30 steps per stage, guidance 3.5 then 7, the `super_imax` guidance schedule, shifted-window patchification, proportional attention, RIFLEx positions, and attention decay. Each prompt also needs a 2+2-step block-average preparation pass. The reported campaign time includes preparation when not cached and the two-stage call; main-stage and preparation times are retained separately in sidecars. Each call reseeds both the generator and the global RNG, including the VAE sampling path, from the prompt id and seed label. Thus seed labels are paired, but the two-stage method does not share an identical latent trajectory with the one-stage methods. The wrapper preserves an upstream scheduler behaviour: assigning `scheduler.config.shift` does not update the scheduler's internal shift in the installed diffusers version, so both stages retain shift 3.0. This is a reproduction caveat, not a newly tuned setting. We report this system-level comparison separately from the common-sampler attention comparisons.

HRDiT. We run the released progressive protocol of the official repository (commit `f3be935`) on FLUX.1-dev in fp16: 30 steps at 1024^2 with guidance 3.5, then 17 steps at 2048^2 with guidance 4.5 and 10 steps at 4096^2 with guidance 6, 57 steps in all, with latent upsampling and the shipped per-head scope plan and attention settings. Its sampling budget therefore differs from the shared protocol, and we report it with the multi-stage recipes. HRDiT's global Torch random state is reset per prompt because its Gaussian blur draws a random width. Its clean time, 108.6 s, is the median of three batch-1 runs on the latency prompts (the first, 144.4 s, includes warm-up). The 191-prompt HRDiT evaluation used the original crop assembly, under which patch FID is 110.7 for HRDiT and 106.5 for SCA. Recomputed from the cached crop features with the corrected assembly of Table 1 (`repro/evaluate_corrected.py`), HRDiT's patch FID is 103.9, against 99.8 for SCA and 103.4 for AReA.

Progressive SCA. `scripts/sample_progressive.py` reimplements HRDiT's released schedule with the stock diffusers FLUX components and changes only the attention (Section 2.5). One 30-step schedule with dynamic shift $\mu = 1.15$ (from the 1024^2 sequence length) serves all stages, and the recorded HRDiT noise levels are reproduced to 10^{-6}. The 2048^2 stage repeats steps 14 to 30 and the 4096^2 stage steps 21 to 30. Each transition decodes the latent, upsamples it bicubically, applies HRDiT's unsharp mask with a blur width drawn from the per-prompt global seed, re-encodes it, and re-noises it with the per-prompt generator; guidance is 3.5, 4.5, and 6.0, and HRDiT's low-band structure guidance and velocity momentum are kept. At 1024^2 both methods use dense attention with native RoPE. At 2048^2 and 4096^2 we remove NTK RoPE, the proportional scale, head pruning, and position alignment, and use SCA with the FLUX.1-dev table at scales 2 and 4, $n_0 = 64$, $t = 8$, and $\sigma_g = 0.89$, which no high-resolution step exceeds. Deliberate differences from HRDiT's runner are bf16 instead of fp16 and a finite-value check at the end of each stage instead of after every step; with bf16 and stock components the 1024^2 images, and hence the layouts,

differ from HRDiT's for the same seed (Figure 12). Reported time is CUDA-synchronized and covers text encoding, all stages, transitions, and the final decode, excluding model loading; per-stage means are 5.1, 12.3, and 26.1 s, peak memory 37.8 GiB. The gate and every other setting were fixed before generation; two development prompts served as a smoke test.

SigMa. The dev-matched profile runs the released code (commit `fbe3a86`) on FLUX.1-dev under the shared sampler: 28 steps, guidance 4.5, bf16, YaRN positions, and the shift capped at its maximum from the target sequence length. SigMa's paper states FLUX.1-dev with guidance 3.5, while its released command line hardcodes FLUX.1-Krea-dev with guidance 4.5; we ran only the profile that matches our protocol. The outputs are finite, fully decoded 4096^2 images, but the images we inspected show a regular lattice of dots, as DyPE's do at 4096^2 on FLUX.1-dev. This may reflect our port or profile rather than the method, so the SigMa row should not be read as the method's best result.

Adapter for HRDiT and SigMa. SigMa runs on the 46-prompt subset and HRDiT on all 191 prompts, both through one adapter with campaign per-prompt CPU generators, resident models, and tiled VAE decoding. These differ from the upstream command-line seed, offload, and decode choices, so the runs are controlled adaptations, not reproductions of the authors' tables. Campaign time includes text encoding, decoding, and finite-value checks after every step, which add synchronization overhead; it excludes model loading and is not a clean latency benchmark.

FLUX.2-dev. The model has 56 attention blocks and 48 heads per block. We reuse the two branches and calibration rule with a native grid of 64^2 tokens, local window 64, and query tile 8. The four-axis positions preserve the time and text axes while coarsening the two spatial axes. Calibration uses 16 prompts at 1024^2, 14 noise bins, and emulated scales 2 and 4; sampling uses the scale-4 table. Both full attention and SCA use 28 steps, guidance 4.0, seed label 0, and a shift fixed to the native sequence length, $\mu = 2.1514$. The pipeline's default extrapolated shift at 4096^2 would be about 11.55; we therefore do not interpret this comparison as a default-pipeline benchmark. The gate is 0.959 so that the first 8 steps use only global attention. The 50 evaluation prompts are the first 50 Aesthetic-4K prompts remaining after development-prompt exclusion. Prompt embeddings are cached for both methods; reported times include denoising and tiled VAE decoding, and exclude text encoding and model loading. At emulated scale 4, fitted per-head relative output error is 0.1790, versus 0.1889 for per-layer weights and 0.1924 for local-only attention; split-half correlation is 0.950 and cross-half error 0.1794. These are calibration diagnostics, not generation ablations of the individual components on FLUX.2.

H Extended results with intervals

Tables 9 and 10 give every metric with its paired interval for the main comparisons, Tables 11 to 13 the ablation intervals, and Table 14 the non-square calibration check of Appendix L; Appendix I pairs the added baselines against SCA. All are populated from archived evaluation CSV files. Per-image means were checked against the corresponding per-image records; distribution-level estimates use archived cached features and corrected reference-row crop assembly (Appendix E). Set-level metrics (FID, KID, their patch versions, and Inception score) are computed on the whole set. The final SCA FID/KID and patch-FID/KID intervals resample reference-image clusters; other FID/KID intervals retain the historical reference-row bootstrap unless a corrected image-cluster interval is available, and Inception-score intervals resample only generated images.

Table 8: **At 4096^2, SCA beats AReA on every per-image metric at about a third of its latency.** Aesthetic-4K with FLUX.1-dev, 191 prompts (100 at 6144^2). HPS, CLIP-IQA, and ARNIQA are $\times 100$; higher is better except latency and FID. Bold: best per size. * / † on full-attention and SCA rows: significantly better / worse than AReA (95% paired interval excludes zero). Latency at 4096^2: seconds per image, batch 1, idle GH200, median of three. ‡Mean campaign time on shared nodes, not clean latency. $^\circ$UltraImage uses its two-stage defaults.

Method	Lat. (s)↓	Image Reward	Aesth.	CLIP	Pick Score	HPSv2	HPSv2.1	FID↓	FID_p ↓	MUSIQ	MUSIQ_c	CLIP-IQA	ARNIQA
4096×4096, scale factors (4, 4)													
Full attention	186.5	$-2.18^\dagger$	$4.51^\dagger$	$15.38^\dagger$	$17.94^\dagger$	$20.75^\dagger$	$7.61^\dagger$	$312.7^\dagger$	252.4	$22.9^\dagger$	$33.2^\dagger$	$49.0^\dagger$	$54.5^\dagger$
DyPE	$196.1^\ddagger$	−0.11	5.73	24.84	20.49	25.52	20.52	206.2	172.7	36.3	50.4	59.5	60.1
SEGA	$190.5^\ddagger$	0.55	6.03	26.93	21.64	27.15	24.37	165.4	126.0	37.4	50.4	66.8	64.8
UltraImage$^\circ$	$308.5^\ddagger$	**1.28**	**6.50**	28.80	**23.32**	**29.97**	**30.80**	142.7	110.6	33.5	41.8	72.7	**72.3**
AReA (released code)	202.4	0.92	6.28	27.67	22.35	28.64	27.20	146.0	103.4	39.4	50.4	67.5	67.3
SCA + band guard	88.1	1.20^*	6.46^*	28.87^*	22.90^*	29.40^*	28.95^*	142.3	**98.8**	41.4^*	61.0^*	70.9^*	70.8^*
SCA (final)	**71.1**	1.23^*	6.45^*	**29.11***	23.09^*	29.77^*	29.71^*	**138.4***	99.8	**45.3***	**61.2***	**72.9***	71.2^*
2048×4096, scale factors (2, 4)													
AReA (released code)	$112.4^\ddagger$	**1.13**	**6.41**	**28.51**	**22.67**	28.38	**27.66**	144.8	**112.8**	51.8	61.6	**74.8**	**73.0**
SCA (final)	**38.1**‡	**1.13**	$6.21^\dagger$	28.21	22.64	**28.55***	27.46	**143.8**	$118.3^\dagger$	**55.9***	**65.9***	$73.5^\dagger$	72.8
2048×6144, scale factors (2, 6)													
AReA (released code)	$150.5^\ddagger$	0.39	**6.15**	26.49	21.51	26.41	23.35	163.6	**106.5**	42.9	53.1	67.3	69.3
SCA (final)	**55.8**‡	**0.66***	$6.00^\dagger$	**27.01**	**21.74***	**26.54***	**23.48**	**143.8***	110.1	**48.4***	**60.6***	**69.0***	**70.2***
6144×6144, scale factors (6, 6), 100 prompts													
AReA (released code)	$507.9^\ddagger$	−0.42	5.80	23.84	20.30	24.95	18.78	216.7	188.6	28.3	44.5	53.8	47.8
SCA (final)	**164.7**‡	**0.84***	**6.24***	**27.12***	**21.89***	**27.48***	**25.16***	**169.9***	**122.4***	**37.5***	**48.3***	**60.1***	**62.6***

Table 9: **All metrics at 4096^2**, 191 prompts, paired against AReA (released code). Δ is method minus AReA with its 95% paired bootstrap interval (1000 draws); **blue** / *orange*: significantly better / worse than AReA. Set-level intervals follow the policy stated after Table 8; corrected patch values without recomputed intervals have no significance marks. ‡Mean campaign wall-clock on shared nodes; Table 5 gives clean latency. Sources: `results/final4k_nb_*`, `final4k_*`, `final4k_full_*`.

Metric	AReA	SCA final	Δ [95% CI]	SCA + band	Δ [95% CI]	Full attention	Δ [95% CI]
ImageReward	0.917	1.229	**+0.312** [+0.240, +0.384]	1.198	**+0.281** [+0.208, +0.358]	−2.178	−3.095 [−3.216, −2.978]
Aesthetic	6.28	6.45	**+0.17** [+0.12, +0.22]	6.46	**+0.18** [+0.12, +0.24]	4.51	−1.77 [−1.87, −1.67]
CLIP score	27.67	29.11	**+1.45** [+1.06, +1.82]	28.87	**+1.20** [+0.79, +1.61]	15.38	−12.28 [−13.10, −11.44]
PickScore	22.35	23.09	**+0.73** [+0.62, +0.86]	22.90	**+0.54** [+0.40, +0.71]	17.94	−4.41 [−4.61, −4.18]
HPSv2 (×100)	28.64	29.77	**+1.12** [+0.95, +1.31]	29.40	**+0.75** [+0.53, +0.99]	20.75	−7.89 [−8.23, −7.54]
HPSv2.1 (×100)	27.20	29.71	**+2.50** [+2.17, +2.88]	28.95	**+1.75** [+1.31, +2.22]	7.61	−19.59 [−20.39, −18.71]
MUSIQ	39.4	45.3	**+5.8** [+4.1, +7.5]	41.4	**+2.0** [+0.0, +3.9]	22.9	−16.5 [−18.0, −15.0]
MUSIQ_c	50.4	61.2	**+10.9** [+9.5, +12.4]	61.0	**+10.6** [+9.2, +12.3]	33.2	−17.1 [−18.8, −15.4]
CLIP-IQA (×100)	67.5	72.9	**+5.4** [+4.0, +6.8]	70.9	**+3.3** [+1.8, +4.9]	49.0	−18.6 [−20.4, −16.7]
ARNIQA (×100)	67.3	71.2	**+3.9** [+3.0, +4.9]	70.8	**+3.5** [+2.4, +4.6]	54.5	−12.8 [−14.3, −11.2]
Inception score	6.42	6.52	+0.09 [−0.47, +0.49]	6.10	−0.33 [−0.88, +0.18]	3.83	−2.60 [−3.14, −1.83]
FID↓	146.0	138.4	**−7.6** [−12.8, −2.9]	142.3	−3.7 [−9.6, +1.0]	312.7	+166.7 [+145.2, +170.9]
KID↓ ($\times 10^3$)	7.11	6.09	−1.02 [−3.12, +1.16]	5.48	−1.63 [−3.84, +0.51]	163.97	+156.86 [+139.17, +174.94]
FID_p ↓	103.4	99.8	−3.6 [−7.4, +1.2]	98.8	−4.7	252.4	+149.0
KID_p ↓ ($\times 10^3$)	11.44	9.17	−2.27 [−4.73, +0.12]	9.67	−1.77	171.16	+159.72
Seconds per image‡ ↓	261.7	71.5	**−190.2** [−203.0, −176.8]	120.0	**−141.7** [−155.8, −127.1]	196.5	**−65.2** [−79.2, −50.5]

Table 10: **All metrics at the non-square sizes**, 191 prompts each, SCA final paired against AReA (released code). Conventions as in Table 9. Corrected patch intervals use reference-image clusters where available; absent intervals are omitted. Sources: `results/final_2048x4096_*`, `final_2048x6144_*`.

	2048 × 4096, factors (2, 4)			2048 × 6144, factors (2, 6)		
Metric	AReA	SCA	Δ [95% CI]	AReA	SCA	Δ [95% CI]
ImageReward	1.133	1.130	−0.003 [−0.079, +0.073]	0.389	0.664	**+0.275** [+0.150, +0.400]
Aesthetic	6.41	6.21	−0.21 [−0.26, −0.15]	6.15	6.00	−0.15 [−0.25, −0.04]
CLIP score	28.51	28.21	−0.30 [−0.70, +0.11]	26.49	27.01	+0.52 [−0.01, +1.07]
PickScore	22.67	22.64	−0.02 [−0.13, +0.09]	21.51	21.74	**+0.23** [+0.05, +0.43]
HPSv2 (×100)	28.38	28.55	**+0.17** [+0.06, +0.28]	26.41	26.54	**+0.13** [+0.00, +0.27]
HPSv2.1 (×100)	27.66	27.46	−0.20 [−0.49, +0.05]	23.35	23.48	+0.13 [−0.27, +0.54]
MUSIQ	51.8	55.9	**+4.1** [+2.7, +5.7]	42.9	48.4	**+5.5** [+3.6, +7.3]
MUSIQ$_c$	61.6	65.9	**+4.3** [+3.3, +5.5]	53.1	60.6	**+7.5** [+6.2, +9.0]
CLIP-IQA (×100)	74.8	73.5	−1.3 [−2.3, −0.4]	67.3	69.0	**+1.6** [+0.2, +3.1]
ARNIQA (×100)	73.0	72.8	−0.2 [−0.9, +0.5]	69.3	70.2	**+0.9** [+0.1, +1.7]
Inception score	6.21	6.65	+0.44 [−0.16, +0.95]	5.36	6.35	**+0.99** [+0.13, +1.22]
FID↓	144.8	143.8	−1.1 [−6.4, +4.3]	163.6	143.8	**−19.8** [−24.4, −11.8]
KID↓ (×10^3)	7.95	7.87	−0.09 [−2.90, +2.97]	18.08	8.46	**−9.62** [−13.94, −5.60]
FID$_p$ ↓	112.8	118.3	+5.5 [+1.9, +9.5]	106.5	110.1	+3.6 [−0.2, +11.3]
KID$_p$ ↓ (×10^3)	14.53	16.48	+1.95 [+0.39, +3.70]	13.66	15.36	+1.70 [−2.22, +6.07]
Seconds per image‡ ↓	112.4	38.1	**−74.3** [−76.6, −72.0]	150.5	55.8	**−94.7** [−96.9, −92.4]

Table 11: **Ablation intervals for Table 3**, 46 prompts at 4096^2, paired against SCA final (2000 bootstrap draws); **blue** / *orange*: significantly better / worse than SCA final. Panels group the variants as in Table 3; the SCA final column gives its absolute values, and the last row of each panel gives patch FID per variant (set-level, no paired interval). Sources: `results/ablnb_*`.

Weight rule after the opening phase				
Metric	SCA final	$\pi = 1$	shared $\pi(\sigma)$	per-layer
ImageReward	1.124	−0.357 [−0.524, −0.207]	−0.090 [−0.173, −0.012]	−0.054 [−0.159, +0.030]
Aesthetic	6.45	−0.21 [−0.35, −0.07]	−0.09 [−0.18, −0.02]	+0.01 [−0.06, +0.07]
CLIP score	29.41	−1.93 [−2.68, −1.20]	−0.38 [−0.78, +0.08]	−0.13 [−0.54, +0.28]
PickScore	22.96	−1.20 [−1.44, −0.99]	−0.38 [−0.51, −0.26]	−0.16 [−0.28, −0.05]
HPSv2 (×100)	29.71	−1.78 [−2.07, −1.50]	−0.96 [−1.19, −0.73]	−0.50 [−0.63, −0.37]
HPSv2.1 (×100)	29.98	−2.86 [−3.35, −2.36]	−1.83 [−2.24, −1.43]	−0.70 [−0.98, −0.41]
MUSIQ	46.1	−3.5 [−6.5, −0.7]	−4.9 [−7.9, −1.9]	+0.7 [−1.6, +3.0]
MUSIQ_c	62.5	−0.3 [−2.5, +1.7]	−10.4 [−12.8, −8.3]	−2.4 [−3.6, −1.1]
CLIP-IQA (×100)	71.3	−4.6 [−6.3, −3.0]	−7.7 [−9.4, −6.0]	−0.9 [−2.0, +0.1]
ARNIQA (×100)	70.5	−1.0 [−2.0, +0.0]	−6.5 [−8.5, −4.8]	−1.5 [−2.3, −0.9]
$\text{FID}_p \downarrow$	168.0	192.2	166.7	167.7
Global branch and opening phase				
Metric	SCA final	$\pi = 0$	mean pool	v0, no gate
ImageReward	1.124	−0.395 [−0.507, −0.295]	−1.518 [−1.847, −1.193]	−1.328 [−1.595, −1.064]
Aesthetic	6.45	−0.76 [−0.87, −0.64]	−0.61 [−0.77, −0.44]	−0.08 [−0.17, +0.01]
CLIP score	29.41	−3.48 [−4.43, −2.59]	−3.59 [−4.76, −2.49]	−1.65 [−2.58, −0.84]
PickScore	22.96	−1.44 [−1.66, −1.22]	−2.60 [−3.00, −2.17]	−2.18 [−2.51, −1.85]
HPSv2 (×100)	29.71	−3.32 [−3.72, −2.94]	−4.04 [−4.66, −3.43]	−1.89 [−2.34, −1.45]
HPSv2.1 (×100)	29.98	−5.66 [−6.35, −4.98]	−9.75 [−11.30, −8.17]	−4.07 [−5.11, −3.04]
MUSIQ	46.1	−3.5 [−6.7, −0.2]	−1.8 [−5.3, +1.9]	−5.6 [−9.0, −2.1]
MUSIQ_c	62.5	−11.7 [−13.6, −9.8]	−4.2 [−8.4, −0.5]	+2.5 [−0.4, +5.5]
CLIP-IQA (×100)	71.3	−27.0 [−30.2, −23.8]	−10.1 [−13.0, −7.1]	**+2.9** [+1.0, +4.9]
ARNIQA (×100)	70.5	−45.1 [−47.6, −42.5]	−4.8 [−6.7, −2.6]	**+4.3** [+2.7, +6.1]
$\text{FID}_p \downarrow$	168.0	237.0	184.6	202.6

Table 12: **Intervals for the band-guard ablations** (Table 25 and further variants), 46 prompts at 4096^2, paired against SCA + band guard, whose absolute values form the first column; **blue** / *orange*: significantly better / worse than this reference. Corrected patch FID is shown without intervals. Sources: `results/abl_*`, `route_*`.

Metric	Reference	no band	no gate	shared $\pi(\sigma)$	per-layer	$\pi = 1$
ImageReward	1.165	−0.041 [−0.109, +0.021]	−0.357 [−0.511, −0.222]	−0.032 [−0.084, +0.015]	−0.024 [−0.055, +0.004]	−0.015 [−0.075, +0.040]
Aesthetic	6.43	+0.01 [−0.05, +0.08]	−0.23 [−0.35, −0.11]	−0.06 [−0.13, +0.00]	+0.03 [−0.01, +0.06]	+0.03 [−0.02, +0.10]
CLIP score	29.01	+0.40 [−0.04, +0.92]	−1.15 [−1.82, −0.47]	−0.09 [−0.53, +0.31]	−0.03 [−0.28, +0.23]	+0.16 [−0.19, +0.53]
PickScore	22.71	**+0.25** [+0.09, +0.41]	−0.86 [−1.13, −0.59]	+0.02 [−0.10, +0.14]	+0.01 [−0.06, +0.08]	**+0.12** [+0.03, +0.22]
HPSv2 (×100)	29.18	**+0.54** [+0.31, +0.75]	−0.49 [−0.87, −0.14]	−0.20 [−0.42, +0.01]	−0.08 [−0.19, +0.04]	**+0.16** [+0.04, +0.29]
HPSv2.1 (×100)	28.81	**+1.17** [+0.71, +1.64]	−1.39 [−2.18, −0.63]	−0.33 [−0.68, +0.01]	−0.01 [−0.19, +0.19]	**+0.55** [+0.30, +0.79]
MUSIQ	43.1	**+2.9** [+0.7, +5.7]	−0.7 [−3.6, +2.4]	+0.5 [−2.5, +4.0]	−0.3 [−1.9, +1.8]	+1.0 [−0.3, +2.6]
MUSIQ_c	62.1	+0.4 [−0.9, +1.6]	−1.0 [−3.4, +1.7]	−5.7 [−7.2, −4.5]	−1.5 [−2.4, −0.7]	**+0.9** [+0.3, +1.4]
CLIP-IQA (×100)	69.6	**+1.7** [+0.3, +3.2]	−0.6 [−2.7, +1.5]	−6.5 [−8.4, −4.7]	−1.2 [−2.1, −0.5]	+0.5 [−0.9, +1.9]
ARNIQA (×100)	69.6	+0.9 [−0.6, +3.1]	+1.6 [−0.4, +4.5]	−4.6 [−7.1, −2.9]	−1.2 [−2.0, −0.6]	+0.6 [−0.1, +1.4]
$\text{FID}_p \downarrow$	166.7	+1.3	+8.9	−4.6	−1.3	+1.6

Table 13: **Band-guard ablations, continued.** Conventions as in Table 12.

Metric	Reference	route 0.5	route 0.3	mean pool	$\pi = 0$	v0
ImageReward	1.165	−0.051 [−0.095, −0.009]	−0.026 [−0.082, +0.024]	−0.848 [−1.083, −0.615]	−0.436 [−0.537, −0.336]	−1.369 [−1.633, −1.113]
Aesthetic	6.43	−0.02 [−0.08, +0.04]	+0.02 [−0.03, +0.07]	−0.39 [−0.54, −0.25]	−0.75 [−0.86, −0.63]	−0.07 [−0.15, +0.03]
CLIP score	29.01	−0.00 [−0.31, +0.28]	+0.13 [−0.23, +0.51]	−2.96 [−4.18, −1.67]	−3.08 [−3.98, −2.16]	−1.25 [−2.02, −0.40]
PickScore	22.71	+0.04 [−0.03, +0.12]	**+0.09** [+0.01, +0.17]	−1.80 [−2.10, −1.49]	−1.19 [−1.41, −0.96]	−1.94 [−2.23, −1.65]
HPSv2 (×100)	29.18	+0.02 [−0.11, +0.15]	+0.11 [−0.01, +0.24]	−2.35 [−2.81, −1.94]	−2.79 [−3.18, −2.45]	−1.35 [−1.74, −0.95]
HPSv2.1 (×100)	28.81	+0.18 [−0.04, +0.40]	**+0.34** [+0.12, +0.58]	−5.81 [−6.96, −4.74]	−4.48 [−5.17, −3.80]	−2.90 [−3.82, −2.01]
MUSIQ	43.1	+1.2 [−0.2, +3.0]	+0.8 [−0.4, +2.5]	−1.8 [−5.5, +1.7]	−0.6 [−4.4, +3.4]	−2.7 [−6.1, +0.7]
MUSIQ_c	62.1	+0.5 [−0.0, +1.1]	**+0.8** [+0.2, +1.4]	−0.8 [−5.0, +3.6]	−11.3 [−13.3, −9.6]	+2.9 [−0.3, +6.2]
CLIP-IQA (×100)	69.6	+0.3 [−1.3, +1.5]	+0.6 [−0.7, +1.7]	−5.4 [−9.3, −1.6]	−25.2 [−29.1, −21.4]	**+4.6** [+2.1, +7.3]
ARNIQA (×100)	69.6	+0.1 [−0.6, +0.8]	+0.6 [−0.1, +1.3]	−1.1 [−3.6, +1.9]	−44.3 [−47.4, −40.7]	**+5.1** [+2.8, +8.1]
$\text{FID}_p \downarrow$	166.7	+1.5	+1.1	+18.7	+70.3	+35.9

Table 14: **Calibration table at 2048×4096**, 46 prompts, paired against SCA final with the scale-4 table; **blue** / *orange*: significantly better / worse than this reference. Sources: `results/ps2_*`.

Metric	SCA, scale-2 table	AReA (released code)
ImageReward	+0.012 [−0.025, +0.050]	+0.027 [−0.141, +0.192]
Aesthetic	−0.00 [−0.03, +0.02]	**+0.29** [+0.16, +0.43]
CLIP score	−0.02 [−0.22, +0.17]	+0.19 [−0.53, +0.91]
PickScore	**+0.05** [+0.00, +0.09]	+0.14 [−0.09, +0.39]
HPSv2 (×100)	**+0.09** [+0.04, +0.13]	−0.19 [−0.44, +0.06]
HPSv2.1 (×100)	**+0.17** [+0.08, +0.26]	+0.39 [−0.21, +1.01]
MUSIQ	+0.0 [−0.5, +0.5]	−3.1 [−7.5, +1.2]
MUSIQ_c	−0.0 [−0.4, +0.2]	−4.1 [−6.7, −1.3]
CLIP-IQA (×100)	−0.6 [−1.2, +0.0]	+1.3 [−0.8, +3.4]
ARNIQA (×100)	−0.2 [−0.6, +0.1]	+0.4 [−1.5, +2.8]

I ADDITIONAL PAIRED INTERVALS

Table 15: **Gains over AReA hold on a second prompt set and two more seeds, calibration size barely matters, and on FLUX.1-Krea-dev SCA matches AReA.** Paired differences at 4096^2 from the stated reference; HPSv2.1 is $\times 100$ and n counts evaluation prompts. Calibration and gate rows compare with default SCA (32 calibration prompts, $\sigma_g = 0.89$). Tables 18, 19, and 16 give the intervals.

Setting	n	Reference	ΔImageReward	ΔHPSv2.1	ΔMUSIQ$_c$
HRGen-1K	200	AReA	+0.642	+4.48	+12.71
Seed 1	46	AReA	+0.305	+2.19	+11.58
Seed 2	46	AReA	+0.355	+2.80	+11.04
4 calibration prompts	46	SCA	+0.012	−0.02	+0.09
8 calibration prompts	46	SCA	+0.010	+0.01	−0.03
16 calibration prompts	46	SCA	−0.007	−0.01	−0.11
Gate $\sigma_g = 0.85$	46	SCA	−0.015	−0.85	−2.04
Gate $\sigma_g = 0.93$	46	SCA	−0.004	−0.15	+0.65
FLUX.2-dev	50	Full attention	+0.456	+1.84	+21.44
FLUX.1-Krea-dev	100	AReA	−0.042	+0.78	+4.61

Per-metric intervals are in Tables 17 and 20 for the added baselines, Table 18 for the prompt-set, seed, and model checks, and Table 19 for the calibration-budget and gate checks. Figure 10 plots four of these metrics for five of the six baselines run on all 191 prompts (HRDiT is in Table 17), signed as SCA minus baseline. Every difference in the tables is method minus the named reference. Positive values favour the method for the ten per-image metrics, and **blue** / *orange* mark intervals that exclude zero in favour of / against the method. These comparisons resample matched prompt ids, with 2000 draws and seed 0; they do not treat multiple crops as independent images. The HRGen-1K sample has 200 prompts, the two extra seeds and sensitivity experiments each have 46, FLUX.2-dev has 50, and the main baseline comparison has 191. Tables use the separately recomputed per-image CSVs; these intervals differ slightly from the original 1000-draw evaluator summaries. No multiplicity correction is applied.

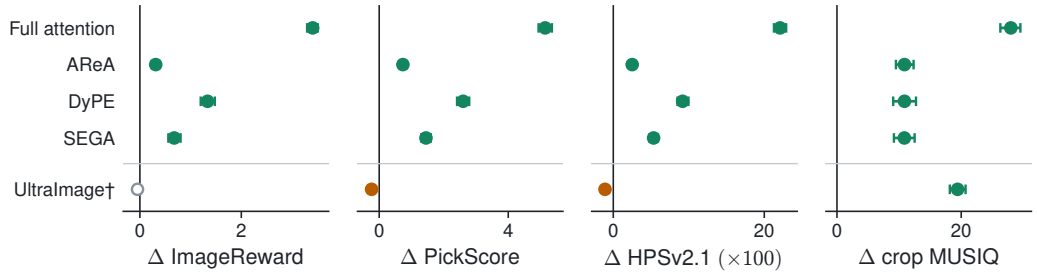

Figure 10: **SCA beats every shared-protocol baseline on all four metrics; against UltraImage the result is mixed.** SCA minus baseline, mean over the same 191 prompts at 4096^2, whiskers 95% prompt-bootstrap intervals (2000 draws, one seed); open markers: interval contains zero; orange: significant loss. †Own two-stage recipe, not an equal-cost comparison.

6144^2 **and FLUX.1-Krea-dev.** Table 16 gives every metric with its paired interval for the 100-prompt 6144^2 comparison and for FLUX.1-Krea-dev, where DyPE and SEGA also trail AReA on ImageReward. The Krea table follows the FLUX.1-dev calibration procedure with 16 prompts at 1024^2 (Appendix F).

46-prompt panel. Table 21 pairs each baseline of Table 2 with SCA on the same 46 prompts and seeds. Per-image intervals use 2000 prompt draws with seed 0; FID, KID, and their patch versions use 1000 draws that resample generated prompts and reference-image clusters. With 46 generated

Table 16: **At 6144^2 SCA beats AReA on 14 of 15 metrics; on FLUX.1-Krea-dev it matches AReA in a third of the time.** Method minus AReA with 95% paired intervals, 100 prompts each; first column FLUX.1-dev at 6144^2, others FLUX.1-Krea-dev at 4096^2 with a Krea-calibrated SCA table. Per-image intervals use 1000 prompt draws; FID, KID, and their patch versions resample prompts and reference-image clusters. Lower is better for FID, KID, their patch versions, and time. **Blue** / *orange*: significantly better / worse than AReA. Time is the campaign mean on shared GH200 nodes.

	6144^2	FLUX.1-Krea-dev, 4096^2		
Metric	SCA	SCA	DyPE	SEGA
ImageReward	**+1.262** [+1.051, +1.450]	−0.042 [−0.118, +0.031]	*−0.282* [−0.424, −0.173]	*−0.110* [−0.189, −0.030]
Aesthetic	**+0.44** [+0.31, +0.55]	**+0.09** [+0.04, +0.15]	+0.05 [−0.02, +0.12]	−0.05 [−0.11, +0.02]
CLIP score	**+3.28** [+2.32, +4.18]	−0.08 [−0.45, +0.27]	*−0.95* [−1.47, −0.49]	*−0.49* [−0.91, −0.07]
PickScore	**+1.58** [+1.31, +1.88]	+0.03 [−0.09, +0.16]	*−0.73* [−0.93, −0.55]	*−0.23* [−0.36, −0.10]
HPSv2 (×100)	**+2.53** [+2.12, +2.94]	**+0.35** [+0.17, +0.53]	*−1.02* [−1.24, −0.80]	*−0.25* [−0.43, −0.09]
HPSv2.1 (×100)	**+6.37** [+5.39, +7.38]	**+0.78** [+0.36, +1.17]	*−1.24* [−1.76, −0.76]	−0.25 [−0.61, +0.09]
MUSIQ	**+9.2** [+6.9, +11.7]	**+1.7** [+0.3, +3.1]	*−2.6* [−4.6, −0.4]	+0.7 [−1.2, +2.8]
MUSIQ_c	**+3.8** [+1.4, +6.2]	**+4.6** [+3.4, +5.7]	**+3.2** [+1.8, +4.6]	+1.3 [−0.1, +2.6]
CLIP-IQA (×100)	**+6.3** [+4.3, +8.4]	+0.2 [−1.2, +1.6]	+0.9 [−0.7, +2.3]	**+1.8** [+0.4, +3.2]
ARNIQA (×100)	**+14.9** [+13.0, +17.0]	**+1.5** [+0.8, +2.0]	*−1.6* [−2.6, −0.7]	−0.4 [−1.3, +0.4]
FID↓	**−46.8** [−58.6, −35.3]	−0.8 [−7.9, +6.2]	+5.0 [−2.8, +16.3]	−2.3 [−10.0, +7.4]
KID↓ ($\times 10^3$)	**−23.20** [−31.77, −16.84]	+0.58 [−2.02, +2.94]	−0.81 [−4.63, +3.26]	−0.85 [−4.00, +1.53]
FID_p ↓	**−66.2** [−70.2, −48.9]	*+8.4* [+4.0, +14.2]	*+7.7* [+5.0, +18.5]	+0.1 [−3.1, +6.7]
KID_p ↓ ($\times 10^3$)	**−56.88** [−68.46, −47.50]	+2.16 [−0.20, +4.89]	+1.32 [−1.44, +4.94]	−0.81 [−2.48, +0.90]
Inception score	+0.45 [−0.06, +0.84]	−0.11 [−0.50, +0.33]	+0.25 [−0.22, +0.69]	+0.09 [−0.43, +0.56]
Time (s)↓	**−343.1** [−349.3, −337.4]	**−144.8** [−148.5, −140.9]	**−24.3** [−28.9, −19.6]	**−20.1** [−25.0, −15.5]

Table 17: **DyPE and SEGA trail SCA on every per-image metric; UltraImage and HRDiT are mixed.** Baseline minus SCA at 4096^2, 191 prompts; **blue** / *orange*: baseline significantly better / worse than SCA. UltraImage and HRDiT use their own multi-stage recipes; HRDiT's set-level metrics are in Table 1.

Metric	DyPE	SEGA	UltraImage	HRDiT
ImageReward	*−1.334* [−1.478, −1.199]	*−0.677* [−0.803, −0.562]	+0.047 [−0.016, +0.111]	**+0.109** [+0.044, +0.175]
Aesthetic	*−0.72* [−0.81, −0.63]	*−0.42* [−0.49, −0.35]	**+0.06** [+0.01, +0.10]	**+0.09** [+0.05, +0.14]
CLIP score	*−4.27* [−4.89, −3.66]	*−2.19* [−2.65, −1.77]	*−0.31* [−0.63, −0.01]	−0.02 [−0.31, +0.29]
PickScore	*−2.60* [−2.79, −2.41]	*−1.45* [−1.60, −1.31]	**+0.24** [+0.14, +0.33]	**+0.36** [+0.27, +0.46]
HPSv2 (×100)	*−4.25* [−4.55, −3.97]	*−2.61* [−2.85, −2.37]	**+0.21** [+0.05, +0.36]	+0.14 [−0.03, +0.30]
HPSv2.1 (×100)	*−9.18* [−9.95, −8.50]	*−5.34* [−5.88, −4.81]	**+1.10** [+0.80, +1.40]	**+1.26** [+0.98, +1.55]
MUSIQ	*−9.0* [−11.1, −6.8]	*−7.9* [−9.8, −6.0]	*−11.7* [−13.6, −9.8]	*−5.0* [−6.7, −3.2]
MUSIQ_c	*−10.9* [−12.7, −9.0]	*−10.8* [−12.5, −9.2]	*−19.4* [−20.7, −18.2]	*−7.7* [−9.0, −6.4]
CLIP-IQA (×100)	*−13.4* [−15.3, −11.6]	*−6.2* [−7.8, −4.6]	−0.2 [−1.3, +0.8]	**+2.8** [+1.9, +3.8]
ARNIQA (×100)	*−11.1* [−12.5, −9.7]	*−6.5* [−7.5, −5.4]	**+1.1** [+0.4, +1.8]	+0.4 [−0.2, +1.0]

Table 18: **The gains hold on HRGen-1K, two more seeds, and FLUX.2-dev.** SCA minus AReA for HRGen-1K and seeds 1 and 2; SCA minus full attention for FLUX.2-dev. All at 4096^2; **blue**: significantly better than the reference.

Metric	HRGen-1K	Seed 1	Seed 2	FLUX.2-dev
ImageReward	**+0.642** [+0.539, +0.755]	**+0.305** [+0.172, +0.458]	**+0.355** [+0.244, +0.470]	**+0.456** [+0.320, +0.598]
Aesthetic	**+0.26** [+0.20, +0.33]	+0.11 [−0.00, +0.23]	**+0.16** [+0.06, +0.25]	**+0.30** [+0.18, +0.42]
CLIP score	**+2.50** [+1.99, +3.03]	**+1.51** [+0.94, +2.12]	**+1.68** [+1.05, +2.33]	+0.38 [−0.38, +1.12]
PickScore	**+1.18** [+1.06, +1.31]	**+0.71** [+0.51, +0.92]	**+0.86** [+0.62, +1.11]	**+0.56** [+0.29, +0.86]
HPSv2 (×100)	**+1.87** [+1.67, +2.07]	**+1.02** [+0.71, +1.31]	**+1.35** [+0.99, +1.71]	**+0.85** [+0.50, +1.21]
HPSv2.1 (×100)	**+4.48** [+3.96, +4.98]	**+2.19** [+1.63, +2.74]	**+2.80** [+2.08, +3.47]	**+1.84** [+1.10, +2.55]
MUSIQ	**+6.7** [+4.7, +8.6]	**+5.0** [+1.0, +9.1]	+3.5 [−0.9, +8.0]	**+16.0** [+11.9, +19.9]
MUSIQ_c	**+12.7** [+11.2, +14.3]	**+11.6** [+9.2, +13.9]	**+11.0** [+8.2, +14.3]	**+21.4** [+18.2, +24.8]
CLIP-IQA (×100)	**+8.1** [+6.5, +9.8]	**+2.2** [+0.0, +4.5]	**+4.0** [+1.7, +6.3]	**+6.6** [+4.7, +8.6]
ARNIQA (×100)	**+7.0** [+5.9, +8.0]	**+3.2** [+1.9, +4.4]	**+3.8** [+2.3, +5.4]	**+7.1** [+5.4, +8.8]

Table 19: **The calibration budget barely matters; a gate at $\sigma_g = 0.85$ lowers seven metrics.** Variant minus default SCA, 46 prompts; **blue** / *orange*: variant significantly better / worse than default SCA. Budgets use the first 4, 8, or 16 of the 32 calibration prompts; the default gate is $\sigma_g = 0.89$.

Metric	4 prompts	8 prompts	16 prompts	$\sigma_g = 0.85$	$\sigma_g = 0.93$
ImageReward	+0.012 [−0.007, +0.029]	+0.010 [−0.004, +0.024]	−0.007 [−0.020, +0.005]	−0.015 [−0.098, +0.064]	−0.004 [−0.097, +0.069]
Aesthetic	+0.01 [−0.00, +0.02]	+0.01 [−0.00, +0.02]	+0.00 [−0.01, +0.01]	−0.19 [−0.26, −0.12]	+0.00 [−0.05, +0.06]
CLIP score	+0.08 [−0.03, +0.19]	+0.05 [−0.04, +0.15]	+0.04 [−0.04, +0.13]	−0.27 [−0.82, +0.26]	−0.64 [−1.03, −0.29]
PickScore	−0.01 [−0.03, +0.01]	+0.00 [−0.01, +0.02]	+0.01 [−0.01, +0.03]	−0.12 [−0.27, +0.02]	−0.22 [−0.38, −0.07]
HPSv2 (×100)	+0.02 [−0.02, +0.05]	+0.02 [−0.00, +0.05]	+0.00 [−0.02, +0.03]	−0.48 [−0.69, −0.28]	+0.02 [−0.16, +0.20]
HPSv2.1 (×100)	−0.02 [−0.07, +0.04]	+0.01 [−0.03, +0.05]	−0.01 [−0.05, +0.04]	−0.85 [−1.22, −0.48]	−0.15 [−0.56, +0.24]
MUSIQ	−0.1 [−0.4, +0.3]	−0.1 [−0.5, +0.2]	**+0.1** [+0.0, +0.3]	−1.9 [−3.5, −0.2]	+0.4 [−1.0, +2.0]
MUSIQ_c	+0.1 [−0.1, +0.3]	−0.0 [−0.2, +0.1]	−0.1 [−0.2, +0.0]	−2.0 [−3.0, −1.1]	+0.6 [−0.4, +1.9]
CLIP-IQA (×100)	+0.2 [−0.1, +0.4]	+0.2 [−0.1, +0.4]	−0.1 [−0.4, +0.1]	−3.2 [−4.8, −1.7]	+0.0 [−1.2, +1.1]
ARNIQA (×100)	+0.1 [−0.1, +0.3]	+0.0 [−0.1, +0.2]	−0.0 [−0.1, +0.1]	−4.0 [−5.9, −2.3]	**+1.8** [+0.7, +3.1]

Table 20: **All three added baselines have significantly higher FID, patch FID, and patch KID than SCA.** Method minus SCA on 191 prompts (negative favours the method); *orange*: significantly worse than SCA. Whole-image FID/KID keep the historical 1000-draw reference-row bootstrap; patch estimates use corrected crops and 1000-draw reference-image cluster intervals.

Metric	DyPE	SEGA	UltraImage
FID↓	+67.8 [+56.9, +79.3]	+27.0 [+21.0, +38.4]	+4.4 [+1.2, +10.5]
KID↓ (×10^3)	+41.49 [+32.04, +52.45]	+4.98 [+0.95, +10.11]	+0.31 [−1.69, +2.12]
FID_p ↓	+72.9 [+59.6, +81.6]	+26.2 [+17.9, +34.0]	+10.7 [+6.5, +14.7]
KID_p ↓ (×10^3)	+49.57 [+40.36, +59.66]	+13.76 [+8.95, +19.22]	+7.48 [+5.43, +9.83]

images, the whole-image distribution metrics are exploratory. Table 22 gives the matching intervals for the 56-step and guidance-7 SCA runs.

Table 21: **On the 46-prompt panel, every shared-protocol baseline trails SCA on all ten per-image metrics, while HRDiT and UltraImage lead on several preference metrics.** Baseline minus SCA (28 steps) at 4096^2 with 95% paired intervals; HPSv2, HPSv2.1, CLIP-IQA, and ARNIQA are $\times 100$, KID and KID_p $\times 10^3$. For FID, KID, their patch versions, and campaign time (seconds per image), lower is better, so negative favours the baseline. **Blue** / *orange*: baseline significantly better / worse than SCA. SigMa uses its dev-matched profile; UltraImage and HRDiT use their own recipes (Appendix G).

	Single pass, shared sampler				
Metric	Full attn.	DyPE	SigMa	SEGA	AReA
ImageReward	−3.338 [−3.532, −3.127]	−1.358 [−1.656, −1.065]	−1.895 [−2.224, −1.582]	−0.655 [−0.894, −0.436]	−0.291 [−0.434, −0.159]
Aesthetic	−2.13 [−2.30, −1.96]	−0.80 [−0.96, −0.63]	−0.93 [−1.11, −0.74]	−0.41 [−0.53, −0.30]	−0.11 [−0.18, −0.03]
CLIP score	−14.52 [−16.26, −12.74]	−4.55 [−5.77, −3.38]	−6.46 [−7.75, −5.22]	−2.05 [−2.78, −1.33]	−1.56 [−2.23, −0.87]
PickScore	−5.23 [−5.63, −4.82]	−2.71 [−3.03, −2.41]	−3.29 [−3.66, −2.92]	−1.40 [−1.70, −1.11]	−0.65 [−0.88, −0.43]
HPSv2	−9.30 [−10.02, −8.61]	−4.42 [−4.93, −3.95]	−5.22 [−5.83, −4.71]	−2.53 [−2.99, −2.10]	−1.02 [−1.33, −0.70]
HPSv2.1	−22.98 [−24.54, −21.24]	−9.74 [−11.01, −8.52]	−11.82 [−13.26, −10.46]	−5.16 [−6.19, −4.16]	−2.22 [−2.89, −1.54]
MUSIQ	−23.0 [−26.2, −19.5]	−9.3 [−13.9, −4.6]	−8.7 [−13.7, −3.5]	−9.6 [−12.8, −6.0]	−6.2 [−10.1, −1.9]
MUSIQ_c	−28.8 [−31.7, −25.9]	−11.6 [−15.2, −8.0]	−12.0 [−15.5, −8.7]	−11.2 [−15.0, −7.7]	−10.5 [−13.2, −7.6]
CLIP-IQA	−24.2 [−26.9, −21.4]	−16.1 [−19.9, −12.3]	−15.6 [−19.7, −11.5]	−7.2 [−10.7, −3.8]	−2.9 [−5.1, −0.7]
ARNIQA	−16.3 [−19.8, −12.8]	−11.2 [−14.0, −8.5]	−11.0 [−13.8, −8.3]	−6.3 [−8.7, −3.9]	−2.2 [−3.7, −1.0]
FID↓	+163.7 [+126.4, +178.2]	+76.0 [+52.8, +105.0]	+83.0 [+59.5, +105.2]	+36.9 [+17.7, +62.0]	+6.3 [−4.0, +17.4]
KID↓	+170.7 [+134.5, +207.8]	+46.9 [+30.8, +70.7]	+58.2 [+42.0, +81.6]	+6.4 [−1.6, +21.9]	+1.0 [−3.3, +5.5]
FID_p ↓	+124.4 [+85.5, +124.2]	+83.5 [+58.8, +100.2]	+87.4 [+62.1, +100.0]	+30.3 [+15.0, +45.2]	+0.4 [−9.8, +9.7]
KID_p ↓	+162.9 [+139.6, +184.8]	+54.4 [+35.4, +76.5]	+67.9 [+50.1, +88.8]	+12.2 [+3.5, +23.8]	−1.2 [−8.1, +4.0]
Time (s)↓	+120.0 [+116.0, +125.0]	+132.6 [+121.3, +145.0]	+113.9 [+113.3, +114.3]	+118.1 [+116.4, +120.0]	+169.4 [+156.2, +186.4]

	Multi-stage, own recipe	
Metric	UltraImage	HRDiT
ImageReward	+0.136 [−0.004, +0.270]	**+0.249** [+0.126, +0.378]
Aesthetic	+0.07 [−0.01, +0.15]	**+0.12** [+0.04, +0.21]
CLIP score	−0.64 [−1.26, −0.05]	−0.03 [−0.66, +0.60]
PickScore	**+0.35** [+0.17, +0.53]	**+0.46** [+0.28, +0.66]
HPSv2	**+0.36** [+0.05, +0.70]	+0.27 [−0.11, +0.66]
HPSv2.1	**+1.56** [+0.85, +2.28]	**+1.62** [+0.92, +2.36]
MUSIQ	−11.2 [−15.1, −7.0]	−5.1 [−8.5, −1.6]
MUSIQ_c	−19.6 [−22.4, −16.6]	−8.4 [−11.9, −5.2]
CLIP-IQA	+0.4 [−1.9, +2.7]	**+2.8** [+0.9, +4.8]
ARNIQA	**+2.1** [+0.3, +4.1]	+1.0 [−0.5, +2.6]
FID↓	+7.4 [−2.0, +17.7]	+10.5 [−0.6, +25.2]
KID↓	+2.0 [−2.6, +7.3]	+0.2 [−3.8, +7.5]
FID_p ↓	+14.2 [+5.5, +22.3]	+10.1 [+2.5, +20.0]
KID_p ↓	+12.0 [+7.1, +17.4]	+5.9 [+1.6, +10.9]
Time (s)↓	+250.4 [+235.6, +269.0]	+34.9 [+31.6, +39.4]

J SAMPLING-BUDGET, GUIDANCE, AND SCHEDULE CHECKS

On the same 46 evaluation prompts, we compare the default 28-step SCA run with 56 steps at guidance 4.5 and with 28 steps at guidance 7 (Table 22). The 56-step run keeps the noise threshold at 0.89; it therefore changes the discretization of both the opening and mixed phases. It improves

ImageReward by 0.097 $[0.039, 0.159]$, HPSv2.1 by 1.01, PickScore by 0.16, and MUSIQ$_c$ by 2.26, but increases corrected patch FID by 7.05 $[1.16, 13.03]$ with reference-image cluster resampling. Increasing guidance to 7 alone lowers HPSv2.1 and PickScore and does not resolve an ImageReward difference. Thus the default schedule does not maximize every metric, and extra sampling can change the preference-quality balance.

Against UltraImage on this subset, 56-step SCA has ImageReward difference -0.038 $[-0.159, 0.078]$, HPSv2.1 difference -0.56 $[-1.21, 0.09]$, and PickScore difference -0.19 $[-0.40, 0.01]$. These intervals span zero and do not establish equivalence. Its MUSIQ advantage is 9.94 $[5.69, 13.98]$ and its MUSIQ$_c$ advantage 21.83 $[18.67, 24.73]$. Campaign means are 71.6 s for default SCA, 140.1 s for 56-step SCA, and 322.0 s for UltraImage on these 46 prompts. This is a sampling-budget sensitivity check, not an equal-FLOP or equal-wall-clock experiment: UltraImage has two resolutions, different guidance, and per-prompt preparation. Against HRDiT on the same prompts (Table 2), the 56-step run stays lower on ImageReward (1.22 against 1.37), HPSv2.1 (30.99 against 31.60), and PickScore (23.12 against 23.42) while taking longer (140.1 against 106.5 s); these are point estimates without paired intervals. Table 23 adds a 42-step run sized to HRDiT's campaign time (110.4 against 106.5 s) and progressive SCA (Section 2.5). The extra steps raise HPSv2, HPSv2.1, ARNIQA, and MUSIQ$_c$ over 28 steps but leave ImageReward unresolved, and HRDiT stays significantly ahead of the 42-step run on ImageReward, PickScore, HPSv2.1, aesthetic score, and CLIP-IQA while trailing it on both MUSIQ scores. Progressive SCA is significantly worse than HRDiT on no per-image metric and better on HPSv2 and ARNIQA; against single-pass SCA it gains ImageReward, PickScore, both HPS scores, CLIP-IQA, and ARNIQA, and loses MUSIQ$_c$. Figure 11 places every setting of the panel by ImageReward and time, and Figure 12 shows HRDiT, progressive SCA, and single-pass SCA on two random prompts.

Table 22: **Doubling the steps raises seven metrics; guidance 7 lowers six.** On 46 prompts, the first two columns compare with default SCA; the last compares 56-step SCA with UltraImage. Per-image differences (first-named setting minus second) and 95% paired intervals, 2000 draws; **blue** / *orange*: the interval excludes zero in favour of / against the first-named setting.

Metric	56 vs. 28 steps	Guidance 7 vs. 4.5	56 steps vs. UltraImage
ImageReward	**+0.097** [+0.039, +0.159]	+0.019 [−0.070, +0.113]	−0.038 [−0.159, +0.078]
Aesthetic	+0.03 [−0.02, +0.09]	*−0.10* [−0.18, −0.03]	−0.03 [−0.12, +0.05]
CLIP score	−0.32 [−0.92, +0.19]	−0.43 [−0.94, +0.01]	+0.32 [−0.43, +1.01]
PickScore	**+0.16** [+0.02, +0.31]	*−0.27* [−0.41, −0.14]	−0.19 [−0.40, +0.01]
HPSv2 (×100)	**+0.34** [+0.13, +0.57]	*−0.30* [−0.51, −0.10]	−0.02 [−0.34, +0.30]
HPSv2.1 (×100)	**+1.01** [+0.59, +1.48]	*−0.74* [−1.08, −0.38]	−0.56 [−1.21, +0.09]
MUSIQ	−1.3 [−3.5, +1.0]	+1.3 [−1.2, +3.7]	**+9.9** [+5.7, +14.0]
MUSIQ$_c$	**+2.3** [+1.2, +3.4]	+1.0 [−0.2, +2.3]	**+21.8** [+18.7, +24.7]
CLIP-IQA (×100)	**+1.5** [+0.2, +2.7]	*−1.5* [−2.9, −0.1]	+1.1 [−1.4, +3.4]
ARNIQA (×100)	**+1.3** [+0.3, +2.4]	*−1.4* [−2.2, −0.7]	−0.8 [−2.9, +1.0]

Distribution metrics of the progressive and 42-step runs. Against single-pass SCA at 28 steps (46 generated images, 168 reference identities, 1000 bootstrap draws; `results/progressive46/base_paired_deltas_distribution.csv`), progressive SCA has significantly higher FID (+7.0 $[0.5, 17.2]$), patch FID (+13.7 $[5.6, 23.2]$), and patch KID (+0.009 $[0.005, 0.015]$), while its KID difference spans zero. For 42-step SCA all four intervals span zero (FID +2.8 $[-4.4, 10.0]$, patch FID +2.3 $[-2.3, 7.9]$).

Table 23: **Against HRDiT, progressive SCA is not significantly worse on any per-image metric; single-pass SCA with 42 steps is.** 4096^2, 46 prompts, seed label 0. Per-image differences (first-named setting minus second) with 95% paired intervals over 2000 prompt draws; **blue** / *orange*: the interval excludes zero in favour of / against the first-named setting. Progressive: HRDiT's schedule with SCA attention (Section 2.5); 42 steps: single-pass SCA sized to HRDiT's campaign time. Intervals that include zero do not establish equivalence.

Metric	Progressive vs. HRDiT	Progressive vs. SCA	42 steps vs. SCA	HRDiT vs. 42 steps
ImageReward	−0.077 [−0.197, +0.030]	**+0.172** [+0.019, +0.305]	+0.062 [−0.003, +0.123]	**+0.187** [+0.073, +0.314]
Aesthetic	−0.051 [−0.129, +0.027]	+0.073 [−0.003, +0.155]	+0.029 [−0.029, +0.082]	**+0.095** [+0.003, +0.187]
CLIP score	−0.36 [−0.83, +0.11]	−0.39 [−0.91, +0.11]	−0.29 [−0.81, +0.15]	+0.26 [−0.40, +0.96]
PickScore	−0.02 [−0.18, +0.14]	**+0.44** [+0.27, +0.62]	+0.06 [−0.06, +0.19]	**+0.40** [+0.19, +0.63]
HPSv2 (×100)	**+0.46** [+0.20, +0.74]	**+0.73** [+0.41, +1.06]	**+0.20** [+0.01, +0.39]	+0.07 [−0.32, +0.48]
HPSv2.1 (×100)	+0.28 [−0.21, +0.77]	**+1.90** [+1.18, +2.61]	**+0.64** [+0.28, +1.06]	**+0.98** [+0.29, +1.70]
MUSIQ	+1.6 [−0.5, +3.7]	−3.5 [−7.0, +0.4]	−0.4 [−2.2, +1.2]	*−4.7* [−8.1, −1.2]
MUSIQ_c	+0.7 [−2.3, +3.8]	*−7.7* [−11.1, −4.5]	**+1.5** [+0.3, +2.6]	*−9.9* [−13.1, −6.7]
CLIP-IQA (×100)	−0.1 [−1.7, +1.4]	**+2.7** [+0.6, +4.7]	+0.5 [−0.6, +1.5]	**+2.3** [+0.5, +4.1]
ARNIQA (×100)	**+3.0** [+2.0, +4.0]	**+4.0** [+2.5, +5.6]	**+1.2** [+0.4, +2.0]	−0.2 [−1.6, +1.3]

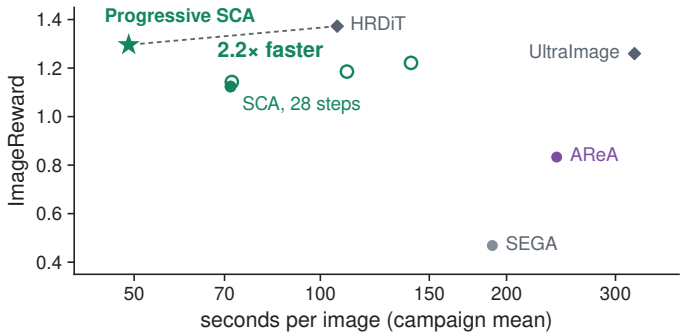

Figure 11: **Progressive SCA reaches HRDiT's ImageReward in less than half its time; extra single-pass steps gain less per second.** 46 prompts at 4096^2, campaign means on shared nodes; diamonds: multi-stage recipes; open circles: SCA at guidance 7, 42 and 56 steps. Below the range: DyPE, SigMa and full attention (Table 2).

Figure 12: **Progressive SCA keeps one large subject, as HRDiT does; single-pass SCA can shrink or repeat it.** Two prompts drawn at random from the 46-prompt panel; same 1:1 512 × 512 crop per row; times in Table 2.

K EXPLORATORY 8192^2 COMPARISON

Table 24 reports all 12 completed prompt pairs at 8192^2 under the shared FLUX.1-dev protocol. SCA has higher ImageReward, CLIP, PickScore, both HPS variants, and ARNIQA, with paired intervals excluding zero. Aesthetic score and the three remaining quality metrics have intervals spanning zero. In particular, MUSIQ, MUSIQ_c, and CLIP-IQA have point estimates favouring AReA. These mixed quality scores and the grid and stripe failures in Figure 23 limit the evidence for reliable 8K generation. With only 12 prompts, this comparison is exploratory; we do not report distributional metrics for this set.

All 8K scores were computed on CPU. HPS uses float32 here, whereas the main GPU evaluation uses mixed precision, so absolute scores across resolutions should not be interpreted as a controlled scaling comparison. Both methods within this table use the same evaluator settings. For native-resolution MUSIQ, attention queries are processed in chunks of 256 while retaining every key and native image patch. This bounds intermediate memory without changing the attention context; pre-trained equivalence checks at three smaller image sizes reproduced the unchunked scores exactly.

Table 24: **Exploratory 8192^2 comparison: SCA is significantly better than AReA on six of ten metrics.** FLUX.1-dev, 12 matched prompts, seed 0. All metrics are higher-is-better. Intervals are paired prompt-bootstrap 95% percentile intervals (2000 draws, bootstrap seed 0); **blue**: significantly better than AReA.

Metric	AReA	SCA	SCA − AReA [95% CI]
ImageReward	−0.376	0.675	**+1.051** [+0.386, +1.754]
Aesthetic	5.64	5.72	+0.08 [−0.26, +0.44]
CLIP score	22.38	25.52	**+3.14** [+0.63, +6.11]
PickScore	20.35	21.27	**+0.92** [+0.24, +1.75]
HPSv2 (×100)	24.60	26.13	**+1.53** [+0.56, +2.66]
HPSv2.1 (×100)	16.59	20.51	**+3.92** [+1.31, +6.85]
MUSIQ	28.6	25.7	−2.9 [−6.8, +0.8]
MUSIQ_c	44.0	40.1	−3.8 [−10.4, +1.6]
CLIP-IQA (×100)	57.1	52.0	−5.1 [−10.5, +0.6]
ARNIQA (×100)	37.3	54.6	**+17.3** [+11.9, +22.6]

L CALIBRATION TABLE AT NON-SQUARE SIZES

At $(s_H, s_W) = (2, 4)$ the two axes have different factors, and SCA uses the table of the emulated scale nearest the larger factor, scale 4. On the 46 ablation prompts at 2048×4096, the scale-2 table changes the paired metrics negligibly (Table 14): HPSv2 +0.09, HPSv2.1 +0.17 (both ×100), and PickScore +0.05 are significant but small, and ImageReward ($+0.01$ $[-0.03, 0.05]$) and the other metrics are within noise. These small changes do not resolve the mixed SCA–AReA results at this size, and we keep the default rule.

M THE LOW-BAND DUPLICATION GUARD

Construction. Let LP_p average-pool a head's image output over $p \times p$ tokens and upsample it bilinearly, and $\text{HP}_p = \text{I} - \text{LP}_p$. The guard replaces the low band of the mix M by that of the global branch:

$$O \;=\; \text{LP}_p\big(O^{\text{glob}}\big) + \text{HP}_p\big(M\big) \;=\; O^{\text{glob}} + \pi_h(\sigma)\,\text{HP}_p\big(O^{\text{loc}} - O^{\text{glob}}\big). \tag{6}$$

The local branch then contributes only its difference from the global branch above the cutoff. The cutoff should lie below $1/(2s)$, where the global branch aliases, and above $1/w$, below which the window cannot resolve; with $w = 64$ and $s = 4$ this leaves periods of 8 to 64 tokens. We used $p = 16$; $p = 8$ and $p = 32$ were worse on four development prompts, and a noise-dependent cutoff behaved like $p = 16$. FreeScale (Qiu et al., 2024) also fuses two self-attention outputs by frequency band, in the opposite direction: it keeps the blurred low band of attention over overlapping crops and takes the high band from full self-attention at the target size. ResDiT also combines global low frequencies and local high frequencies, using patch-wise FFT masks (Ma et al., 2025). Our optional guard uses pooled global attention and preserves the fitted per-head weight on the local-minus-global high-band residual. Its global-low/local-high construction is therefore not itself novel.

History: what it fixes and what it costs. On the development prompts, the band alone (no opening phase) fixed the boats but left extra statues and faces on the Moai and woman-and-bear prompts; the opening phase alone got the main subject right, but small extra figures appeared late on some prompts; both together gave one subject on the boats, Moai, and woman-and-bear prompts. This was our first full configuration. On 46 prompts, however, removing the guard improves HPSv2 (+0.54), HPSv2.1 (+1.17), PickScore (+0.25), CLIP-IQA (+1.7), and MUSIQ (+2.9), all significant, leaves the other metrics within noise, and saves 17 s per image on an idle GPU. The guard also hid what the per-head mix contributes. With it (Table 25), local only matched the per-head mix within 2% on preference metrics, because the guard replaces the low band of every head's output with that of the global branch whatever the head's weight, so it injects global low frequencies even at $\pi = 1$; without the guard local only loses on nearly every metric (Section 4.2). Likewise, with the guard one shared weight and per-layer weights matched the per-head table on preference metrics and lost only on no-reference quality, and hard per-head routing (10 to 17% of heads global at threshold

Table 25: **Ablations around the band-guard configuration at 4096^2**, 46 prompts. The reference row gives absolute values for SCA + band guard (band 16); every other row changes one thing, keeps the guard unless stated, and reports its paired difference to the reference. Marks and units as in Table 3; corrected patch FID is shown without intervals.

Variant	Mix err.	Image Reward	Aesth.	HPSv2	HPSv2.1	Pick Score	CLIP	CLIP-IQA	ARNIQA	MUSIQ	MUSIQ$_c$	FID$_p$ ↓
SCA + band guard (reference)	0.200	1.17	6.43	29.18	28.81	22.71	29.01	69.6	69.6	43.1	62.1	166.7
no band (**= SCA final**)	0.200	−0.04	+0.01	**+0.54**	**+1.17**	**+0.25**	+0.40	**+1.7**	+0.9	**+2.9**	+0.4	+1.3
no gate	0.200	−0.36	−0.23	−0.49	−1.39	−0.86	−1.15	−0.6	+1.6	−0.7	−1.0	+8.9
one shared $\pi(\sigma)$	0.230	−0.03	−0.06	−0.20	−0.33	+0.02	−0.09	−6.5	−4.6	+0.5	−5.7	−4.6
per-layer $\pi_\ell(\sigma)$	0.223	−0.02	+0.03	−0.08	−0.01	+0.01	−0.03	−1.2	−1.2	−0.3	−1.5	−1.3
local only ($\pi = 1$)	0.261	−0.01	+0.03	**+0.16**	**+0.55**	**+0.12**	+0.16	+0.5	+0.6	+1.0	**+0.9**	+1.6
hard routing, threshold 0.5	n/a	−0.05	−0.02	+0.02	+0.18	+0.04	−0.00	+0.3	+0.1	+1.2	+0.5	+1.5
hard routing, threshold 0.3	n/a	−0.03	+0.02	+0.11	**+0.34**	**+0.09**	+0.13	+0.6	+0.6	+0.8	**+0.8**	+1.1
mean pooling	n/a	−0.85	−0.39	−2.35	−5.81	−1.80	−2.96	−5.4	−1.1	−1.8	−0.8	+18.7

0.5, about 6% at 0.3) stayed within 2% of the mix. The guard is an option for prompts where late duplication is visible, not part of the default method.

N Compute and reproducibility

Compute. Generation and calibration used single NVIDIA GH200 GPUs on a shared cluster; the verified timing logs identify the GH200 120 GB model. Some metric evaluations ran on CPUs. The surviving logs do not establish the complete calibration runtime. Generation at 4096^2 takes about 71 s per image for SCA, 262 s for AReA, and 197 s for full attention on shared nodes (campaign means); the main comparisons, ablations, and non-square sizes comprise more than 2000 generated images. Surviving worker logs establish a lower bound of 123.3 GH200 worker-hours through 2026-10-17 06:59 UTC, including model loading, calibration, evaluation, and failed attempts. The bound is the union of 581 closed task intervals per physical node, with timestamp precision treated conservatively; it measures occupied worker time, not GPU kernel activity. Missing historical logs, unmatched starts, and ongoing work prevent a complete campaign total. Overlapping scheduler allocation time and image-sidecar durations are not added to this bound.

Reproducibility. The code (attention processor, calibration, sampling, evaluation wrappers, AReA wrapper, and tests), the calibration table `k1_1024.npz`, the per-image metric files, and the per-comparison summary and paired-delta CSV files behind every table will be released. Main-text metric values and paired intervals are derived from those CSV files. Each SCA generation task ran from a frozen copy of the code named in the task. Per-image sidecar files record the sampling settings and, where available, a code fingerprint; the inherited Krea baseline records do not establish one source revision for every image.

O Broader impact

SCA shortens generation at 4096^2 relative to full attention under our measured protocol, which helps applications such as print, design, and scientific visualization. Energy use remains unmeasured. The same reduction lowers the cost of producing high-resolution synthetic images that could be used to mislead. SCA adds no capability beyond the base model's and does not change its safety filters or watermarking; the risks and mitigations of the base model apply unchanged.

P QUALITATIVE GALLERIES

Figure 13 shows curated SCA outputs. The other galleries, Figures 14 to 23, are uncurated unless stated: prompts are drawn at random from the evaluation prompts, excluding the development prompts and those shown in the main text, and the selection seed, source hashes and crop coordinates of every gallery are archived in its `provenance.json`. Unless a caption says otherwise, images are FLUX.1-dev outputs at seed 0 with 28 steps and guidance 4.5. Every crop, here and in Figures 3 and 12, shows native pixels of the outlined window, which is placed automatically at the highest edge energy averaged over the methods shown, never by hand. Figure 18 uses FLUX.2-dev with guidance 4.0 and a FLUX.2 calibration table. Figure 22 uses the FLUX.1-Krea-dev transformer with FLUX.1-dev text encoders and VAE and a Krea-calibrated SCA table; DyPE and SEGA come from the recorded Krea campaign, in which seven SEGA outputs were regenerated from an archived runtime because their original source snapshots could not be recovered, and the inherited baselines span several runtime revisions. In Figure 20, UltraImage runs its two-stage defaults (30+30 steps, guidance 3.5 then 7), so that comparison is not equal-cost.

Figure 13: **SCA keeps layout and fine detail across three aspect ratios without training.** Curated SCA outputs at 4096^2, 2048×4096 and 2048×6144, picked by eye for variety; insets are 1:1 crops of the outlined windows.

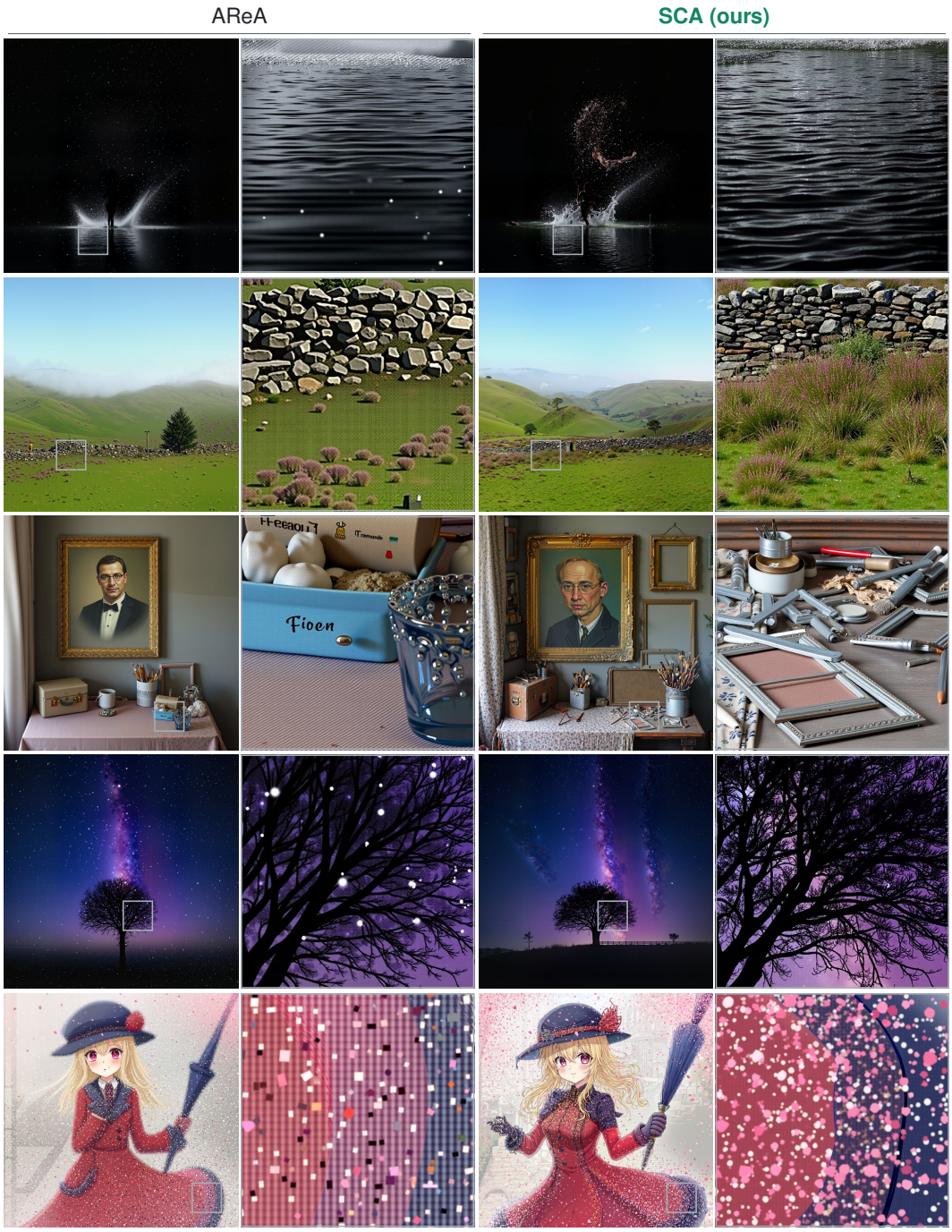

Figure 14: **On uncurated prompts at 4096^2, SCA keeps the fine detail that AReA blurs or tiles.** The first five of 32 prompts drawn at random (seed 0); after each image, a 1:1 crop of its outlined window.

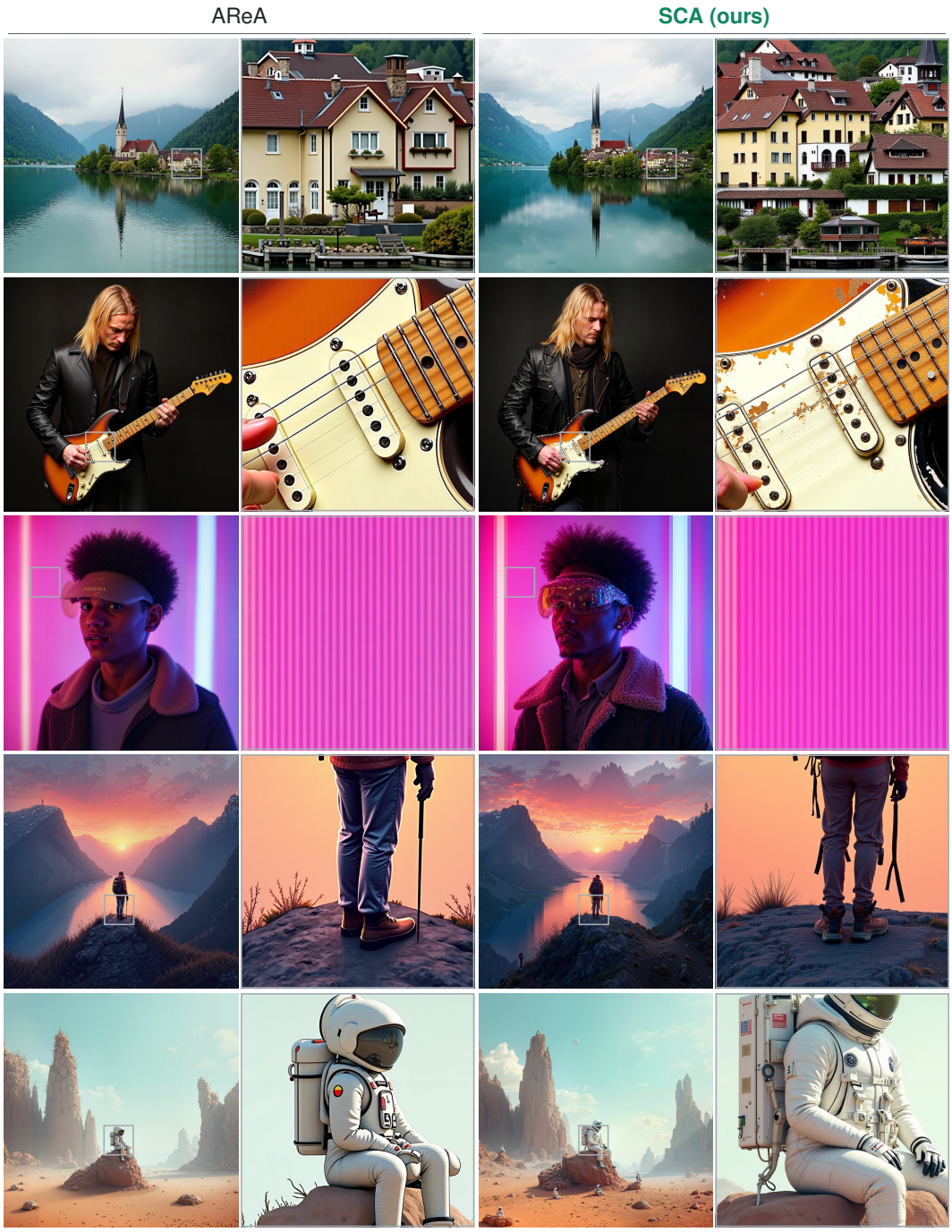

Figure 15: **Uncurated samples at 4096^2, continued.** Prompts six to ten of the same draw; protocol as in Figure 14.

Figure 16: **At 4096 × 2048 both methods keep the layout; SCA resolves finer texture.** The first two of six prompts drawn at random; AReA above, SCA below, each followed by a 1:1 crop of the outlined window.

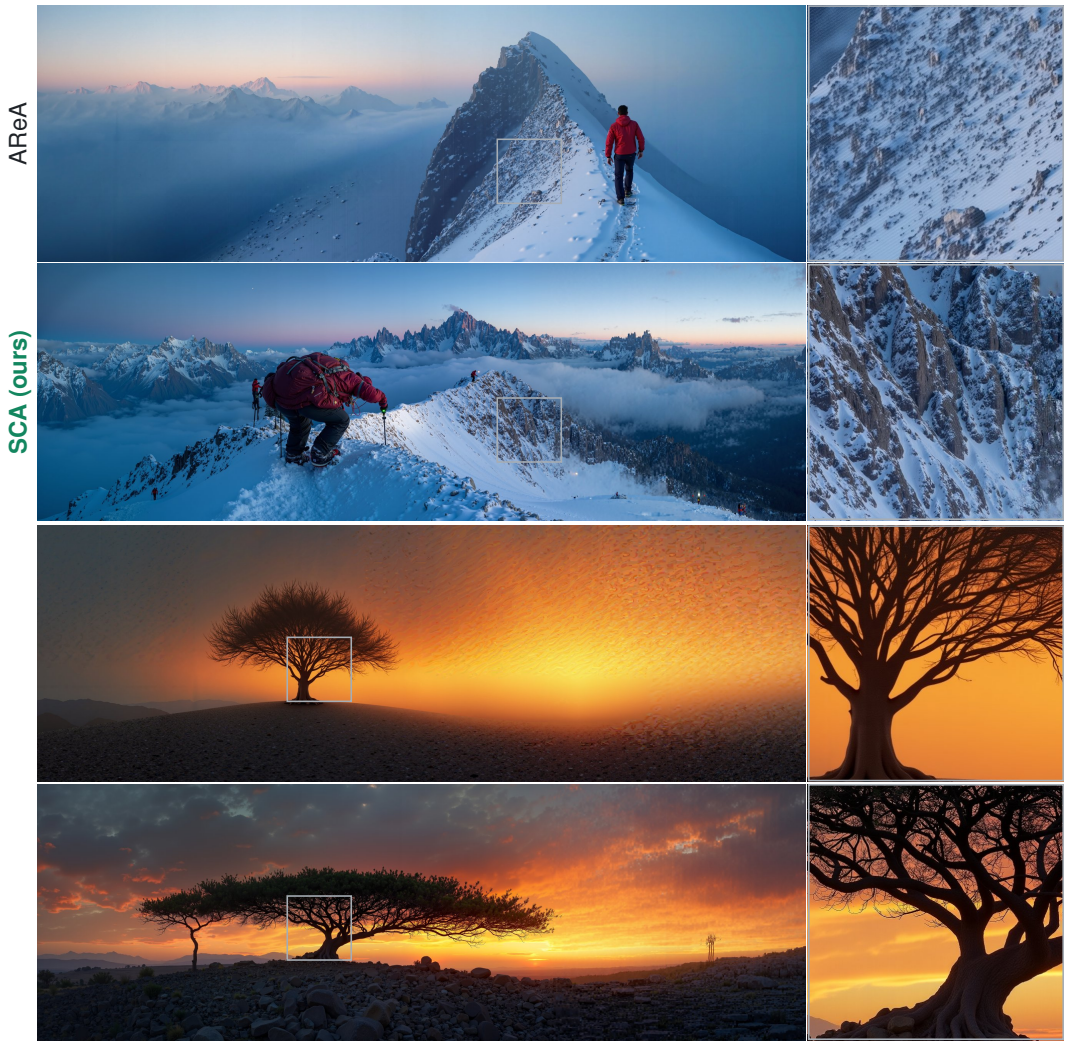

Figure 17: **At 6144×2048 both methods keep the layout; SCA resolves finer texture.** The first two of six prompts drawn at random; AReA above, SCA below, each followed by a 1:1 crop of the outlined window.

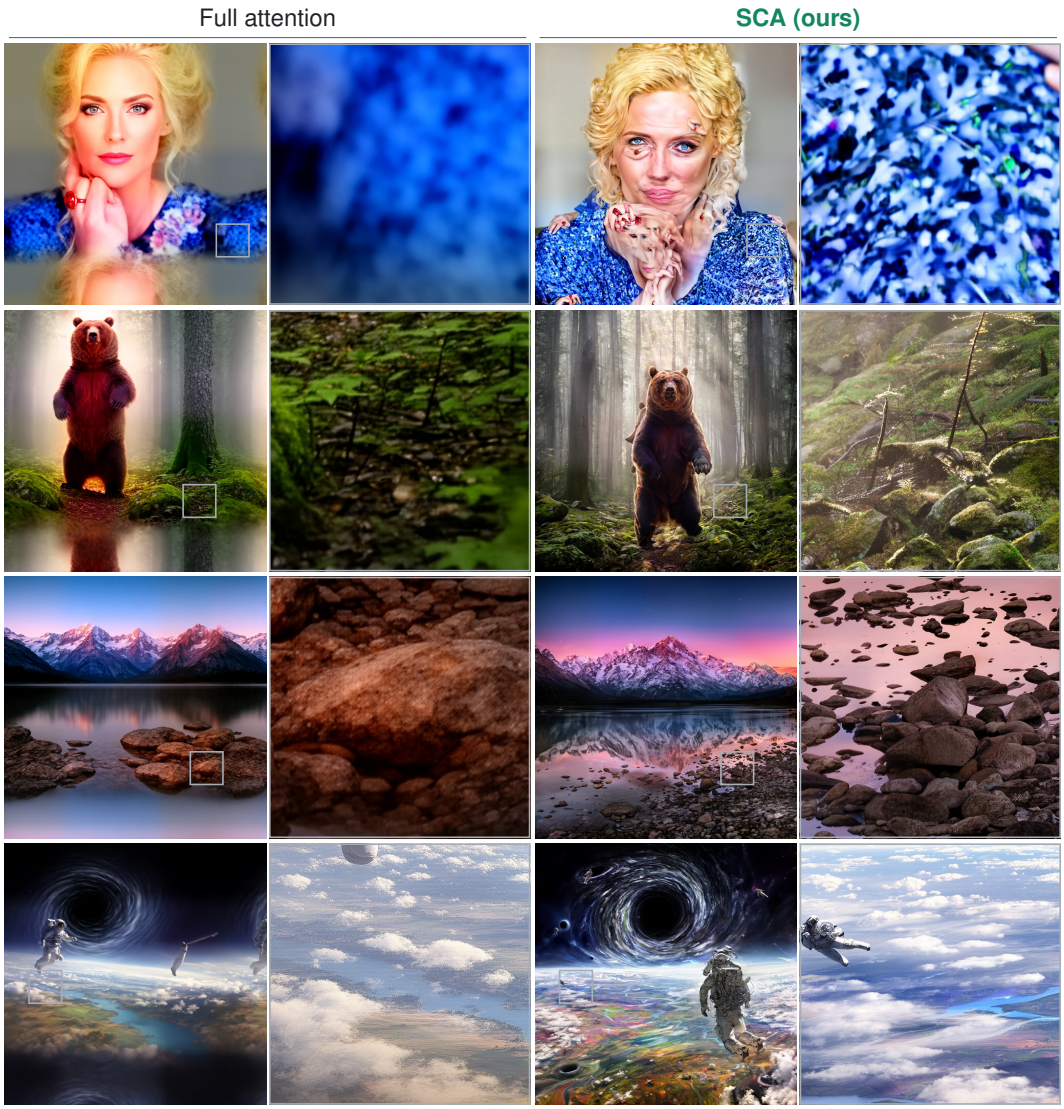

Figure 18: **FLUX.2-dev at 4096^2: SCA can still distort face geometry and fine texture (first row).** Four of 50 paired prompts drawn at random (seed 17); after each image, a 1:1 crop of its outlined window.

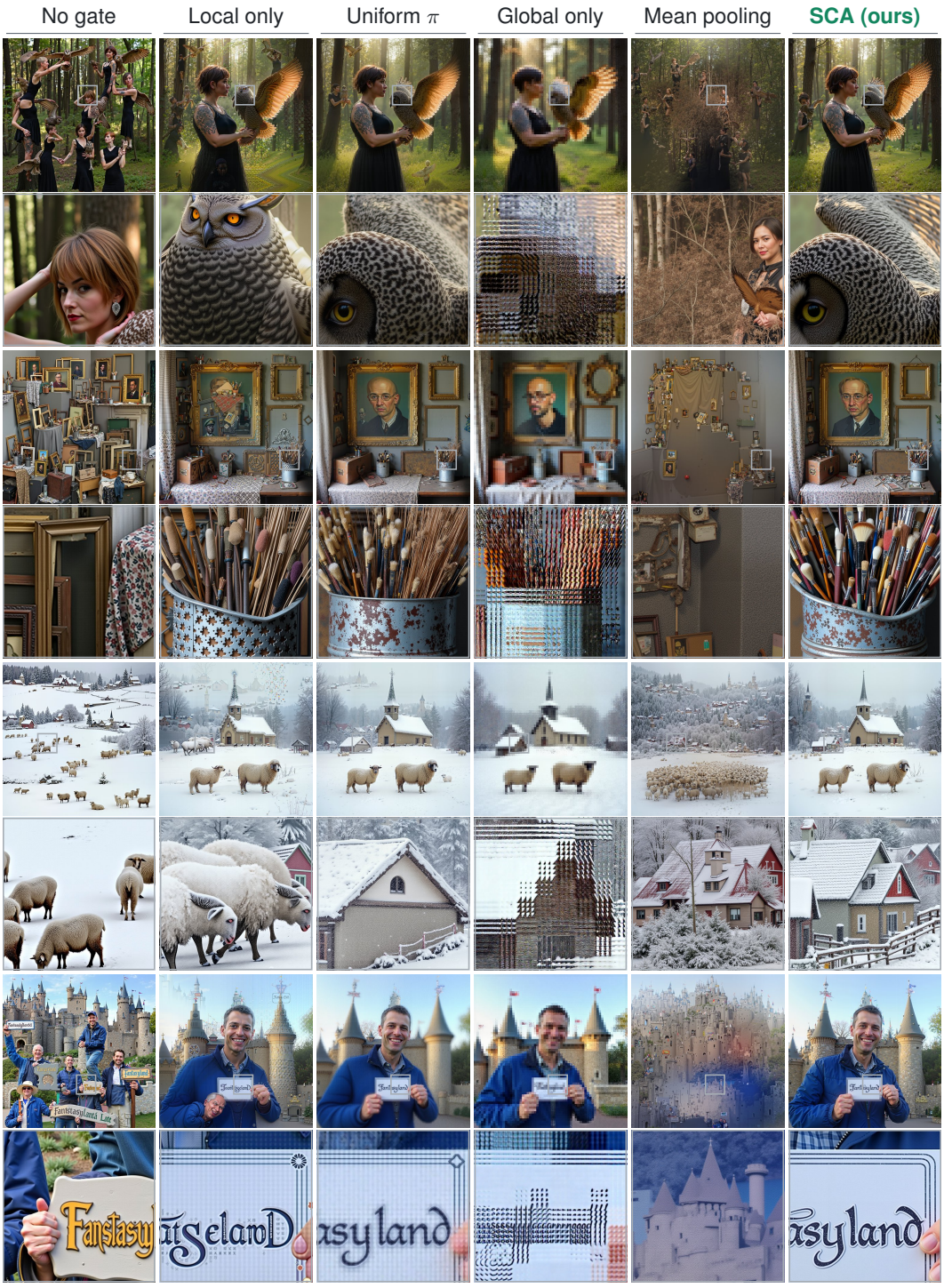

Figure 19: **Without the gate the subject repeats; global only leaves the pooled branch's coarse grid; mean pooling loses the layout.** Four ablation prompts at 4096^2 (Table 3); full images above 1:1 crops of one window.

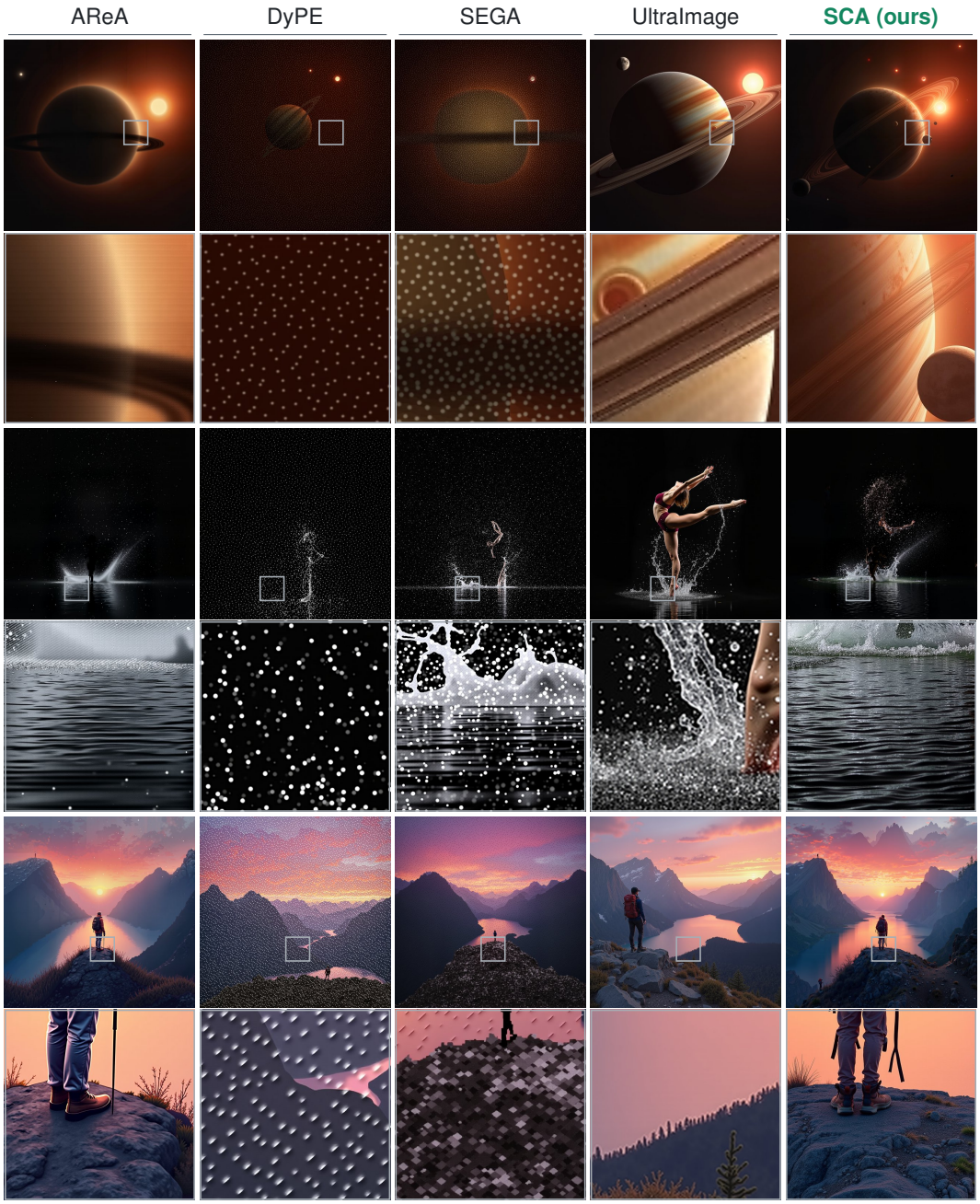

Figure 20: **DyPE and SEGA leave speckle at 4096^2; UltraImage and SCA keep clean detail.** Three of 191 prompts drawn at random (seed 17); full images above 1:1 crops of one window. UltraImage runs its two-stage defaults.

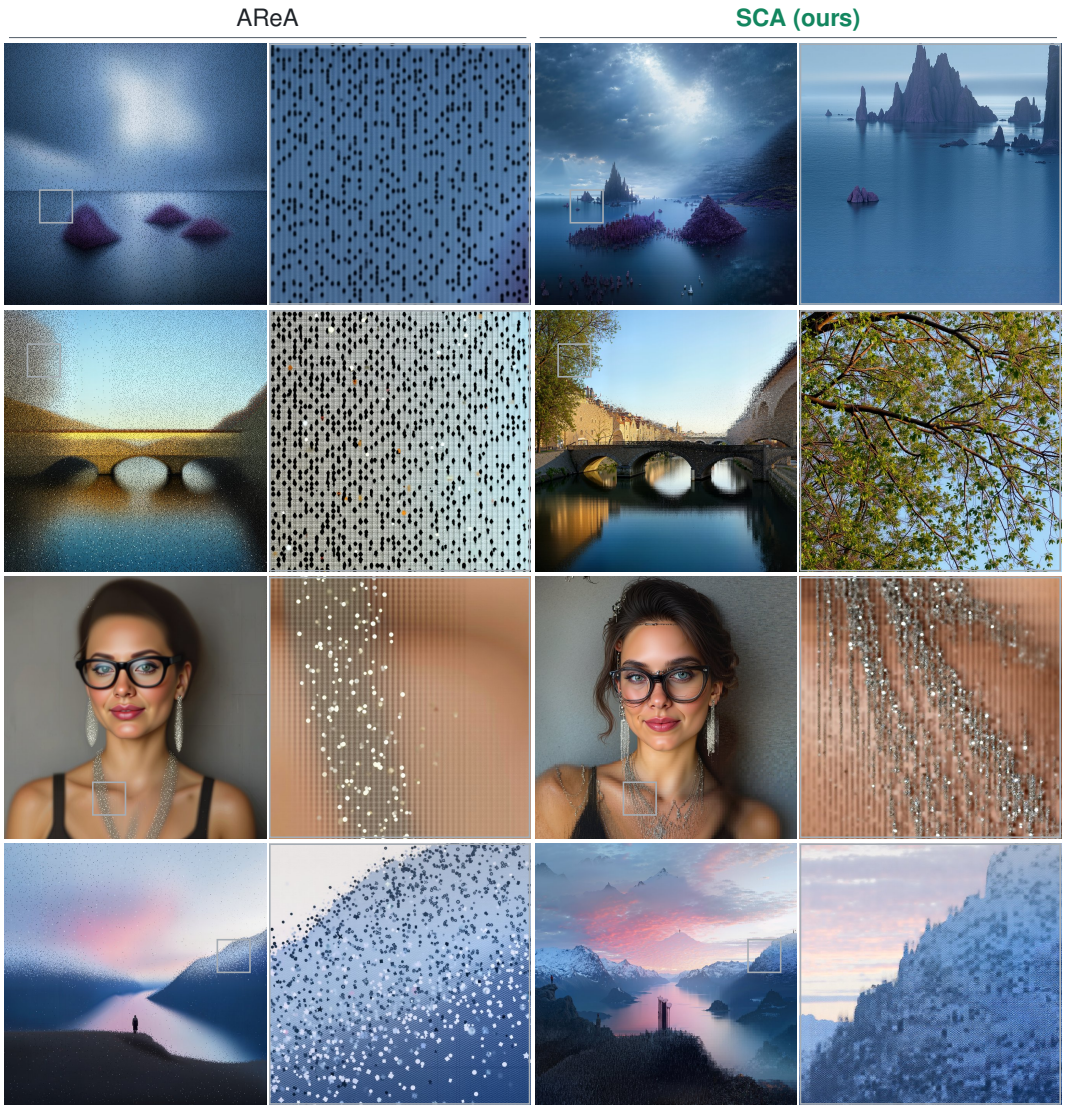

Figure 21: **At 6144^2 AReA's crops dissolve into speckle or streaks; SCA keeps recognisable detail.** Four of 100 prompts, drawn at random; after each image, a 1:1 crop of its window; not evidence of general quality.

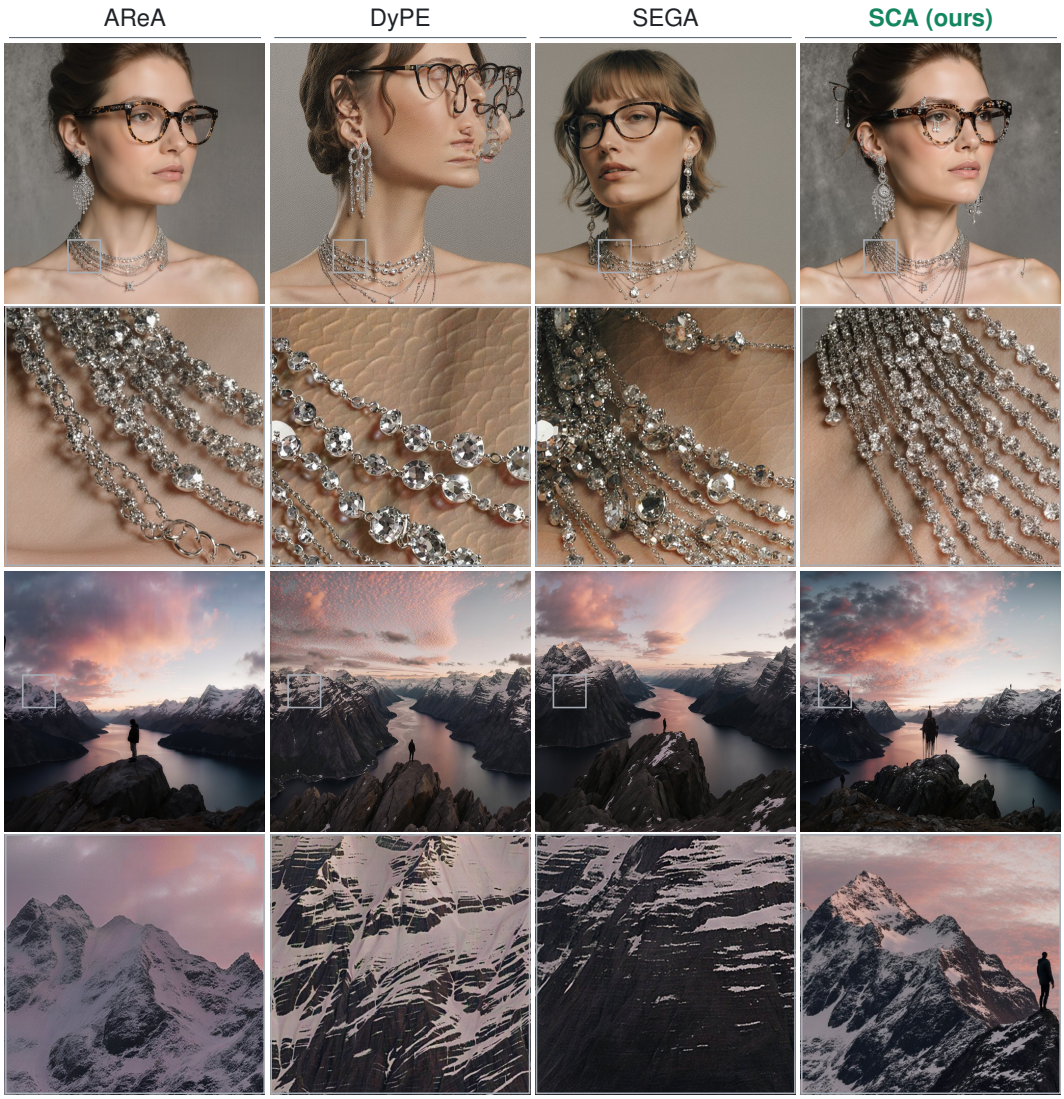

Figure 22: **On FLUX.1-Krea-dev at 4096^2 the four methods are visually close.** Two of 100 four-method prompts drawn at random; full images above 1:1 crops of one window. SCA uses a Krea-calibrated table.

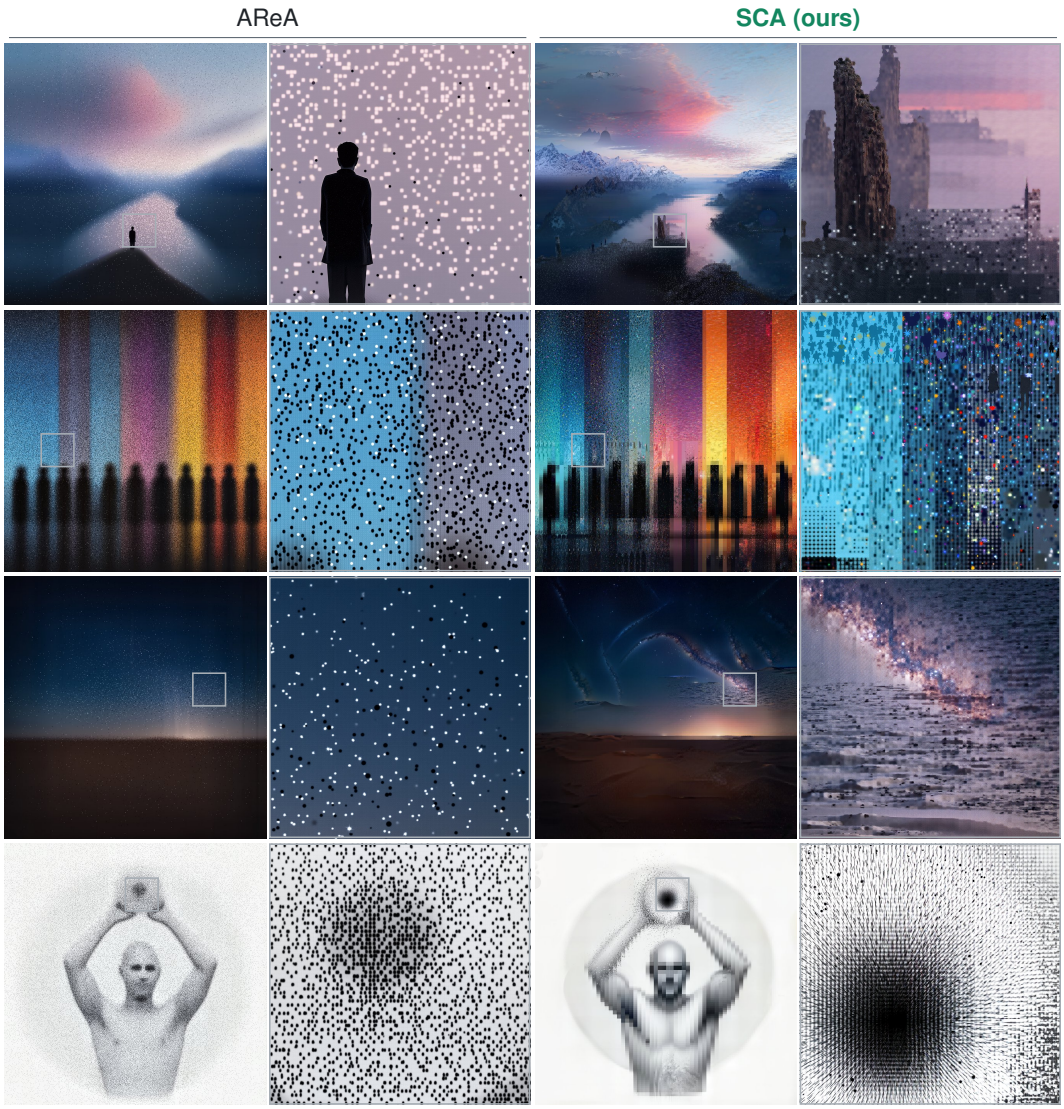

Figure 23: **At 8192^2 grid and stripe artifacts remain in both methods.** Four of 12 paired prompts drawn at random; after each image, a 1:1 crop of its outlined window. Exploratory; not evidence of general 8K quality.

Q FAILURE CASES

Figure 24 shows the duplication that windows cause without the gate, and local errors that SCA still makes.

Late duplication. With the final method, small extra figures can appear on some prompts in the last 20 steps, when the local branch dominates most heads. The band guard suppresses some of them but costs preference scores.

Articulated subjects. The underwater-surfer development prompt keeps doubled limbs in every variant; AReA's image shows a large translucent ghost.

Small factors. At 2048×4096 the window covers half the image height, so the local branch alone already sees most of the image along that axis, and the per-head mix gains little over AReA's shrinking windows; aesthetic score and corrected patch FID are significantly worse.

Larger scales and another backbone. The gallery sampled from the completed 12-prompt 8192^2 run still shows grid and stripe artifacts in both methods (Figure 23); this exploratory set does not establish general 8K quality. In FLUX.2-dev, improved aggregate metrics coexist with distorted facial geometry and fine texture in the randomly selected portrait example (Figure 18).

Text-image offsets. The local branch keeps out-of-range text-image offsets (Remark 1).

Metrics. No-reference quality metrics reward sharp repeated subjects (v0 scores higher on CLIP-IQA and ARNIQA), and FID against 195 references at 299^2 resolution sees little of the 4K detail; we therefore report every metric and the qualitative galleries.

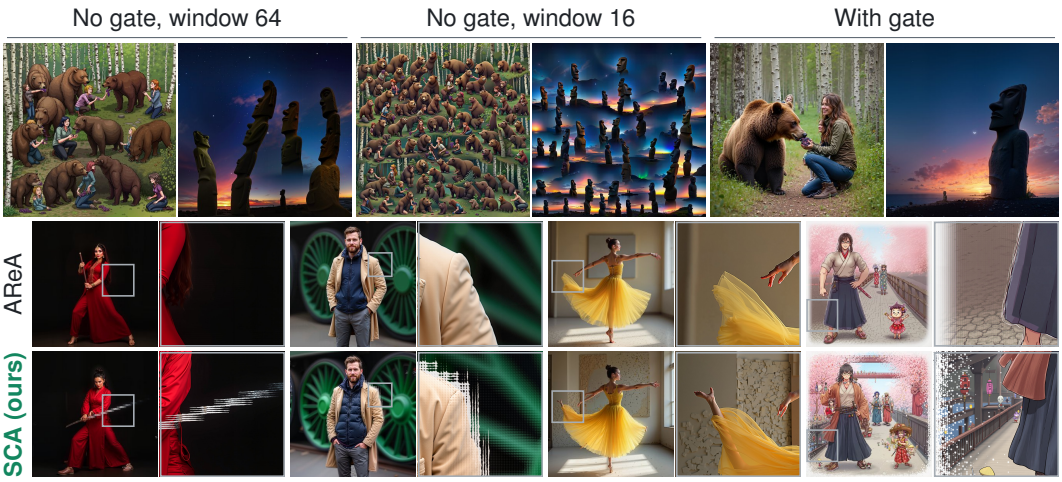

Figure 24: **Without the gate, windows duplicate the subject; SCA still makes local errors.** Top: two development prompts. Below, 1:1 crops of outlined windows: staircase sword, blocky patch, foot for an arm, border speckle; AReA composes differently.